citi bike

Which Citi Bike stations will run out of bikes tomorrow morning?

Predicting shortages to plan the overnight rebalancing in New York City

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Business Analytics Using Python · HEC Paris
Data from August 2025 to August 2026



Each dot is one of the 2,318 Citi Bike stations

AGENDA

Which stations will run empty tomorrow morning, and should the trucks give priority to commuter or leisure stations?



The problem and the data

01 **Problem**
when stations run empty, and what riders say

02 **Data**
four sources: clean, aggregate, sample

Understanding the network

03 **Stations**
four types of stations · PCA, k-means

04 **Demand**
what drives a station's daily rides · linear regression

05 **Riders**
members and casual riders · classification, KNN

Predicting tomorrow

06 **Prediction**
empty tomorrow? · XGBoost, simple rule, truck plan

07 **Robustness**
does the model work in another season?

What to do

08 **Conclusions**
results, recommendations, limits

Appendix

Data: Citi Bike trips and station status, New York City, August 2025 to August 2026

01

PART 1 OF 4

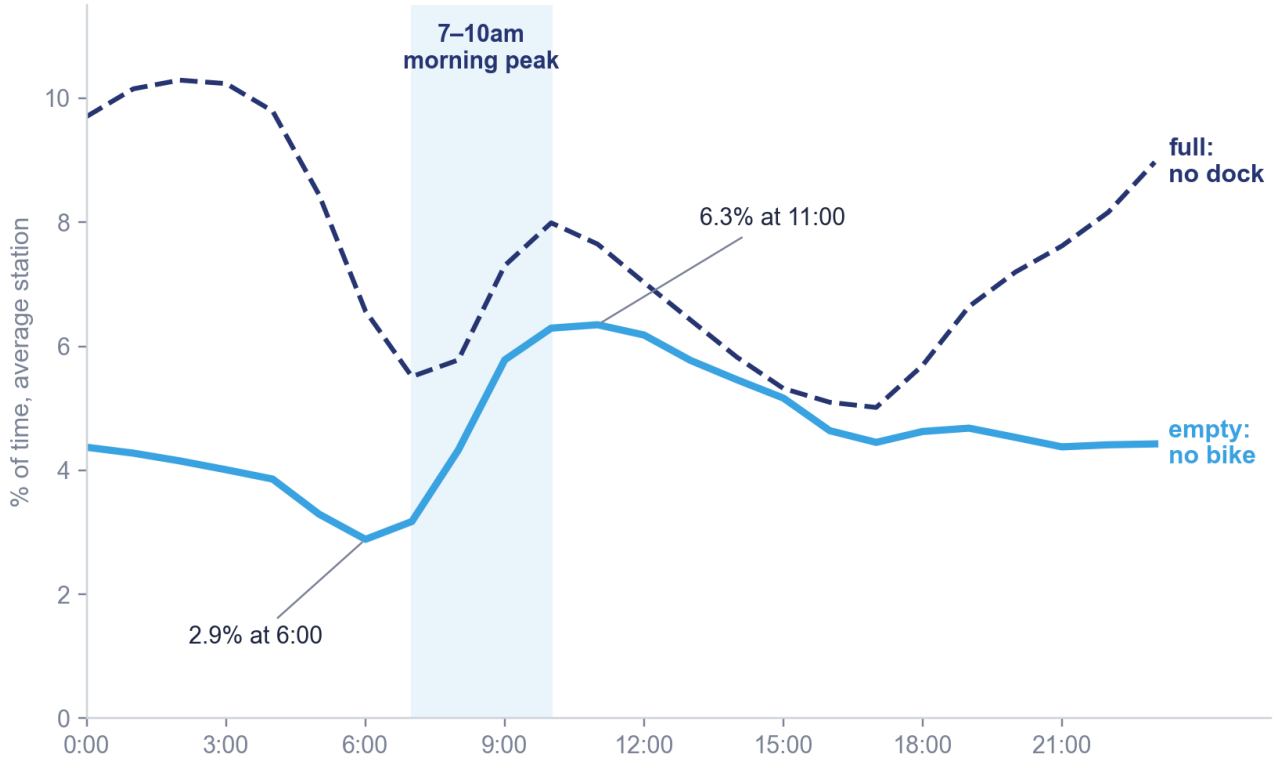
The problem and the data

When and where stations run empty, why it matters, and the data we used.



From 7am, more and more stations have no bike to rent

Share of time an average station is empty (no bike) or full (no free dock), by hour of a weekday.[1]



Shortages start at the morning peak

The share of time a station is empty more than doubles between 6 and 11am. Offices fill up at the same time.

An empty station is a lost ride

The commuter takes another mode of transport, and riders complain about it (next slide).

Trucks have to plan the evening before

Bikes are moved mostly at night, so the list of stations to refill must be ready the evening before.

Riders are also a business question

Casual riders make 18% of rides. Knowing which of them ride like commuters tells Citi Bike whom to offer a membership.

1 Share of the 2-minute snapshots with 0 bikes (empty) or 0 free docks (full), averaged over the 2,318 stations, weekdays, Aug 2025 – Aug 2026. Station status recorded by E. Rosenthal and published on Kaggle.

One App Store review in five is about empty or full docks

Three reviews as examples, and the main topic of 500 App Store reviews found with a topic model (LDA).[1]

> *Downtown Manhattan docks are constantly full … Defeats entire purpose.*

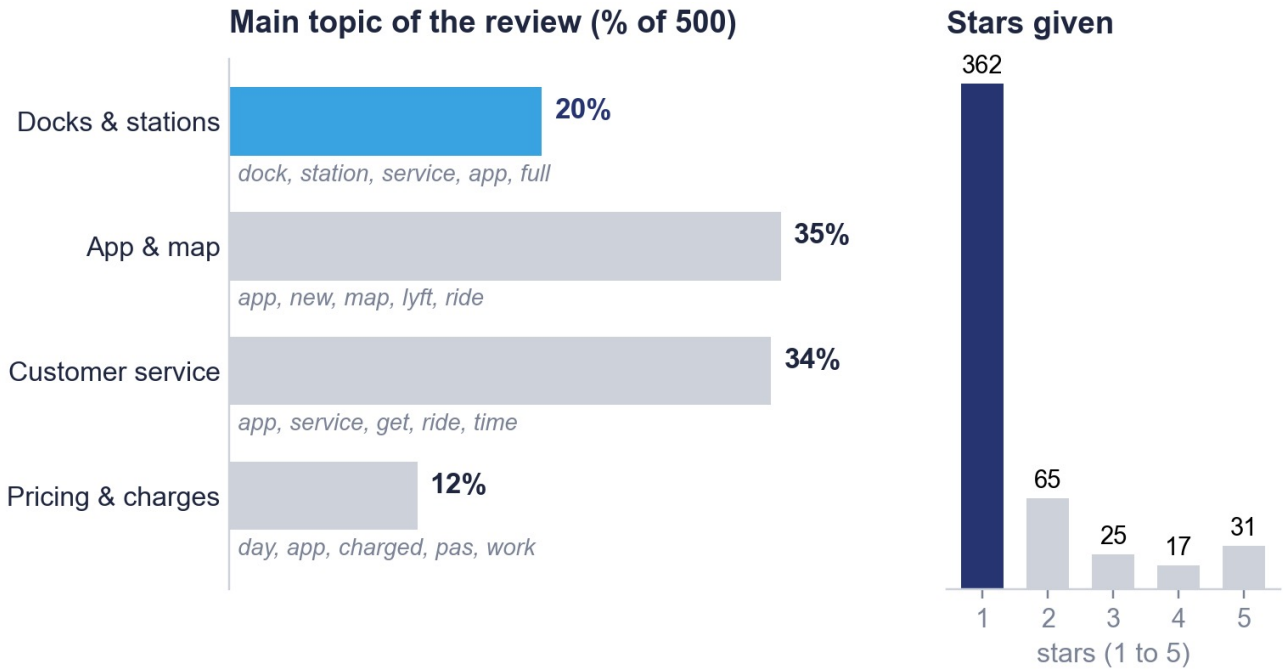
★☆☆☆☆ March 2022

> *It's insane how many of the bikes or stations are constantly broken, and how "unbalanced" they are.*

★☆☆☆☆ June 2026

> *Lots of times there are no docks available and lots of times the bikes are unusable. Still a fast way to get around.*

★★★☆☆ September 2026



Written reviews come mostly from unhappy users: 362 of 500 give one star, while the app's overall rating is 4.5.[2]

1 gensim LdaModel on the 500 written reviews (US App Store, Feb 2021 – Sep 2026), after removing stop words and punctuation and lemmatising; four topics chosen by coherence. Main topic = topic with the highest probability.
2 Written reviews: average 1.58 stars, 362 of 500 give one star. Overall App Store rating of the app: 4.5 out of 5 (10,193 ratings, October 2026), since most users rate without writing.

Four public sources, cleaned and aggregated by station and day

The raw data we downloaded,[1] and the three steps that turn it into the tables used by the models.

Trips
Citi Bike System Data (official)

50.8 M rides · 13 columns

Aug 2025 – Aug 2026

start and end time and station, coordinates, bike type, member or casual

Station status
Kaggle, E. Rosenthal

1,063 M snapshots · 23 columns

Jan 2025 – Sep 2026, every 2 min

bikes and free docks, capacity, in service, station name, time

Weather
Meteostat, Central Park

638 days · 11 columns

Jan 2025 – Sep 2026, daily

average temperature, rain, wind

App reviews
Apple App Store

500 reviews · 7 columns

Feb 2021 – Sep 2026

stars, title, text, app version, date

1 Clean

0.4% of the rides removed (204,183)

- one rule per problem: no station, longer than 3 hours, duplicates, wrong times
- status: 643 M of 1,063 M snapshots kept

2 Aggregate and join[2]

851,613 station-days, 2,318 stations

- share of 7–10am and 4–8pm empty or full
- rides joined by station name, weather and calendar by date

3 Sample

202,423 rides, 0.4% stratified sample

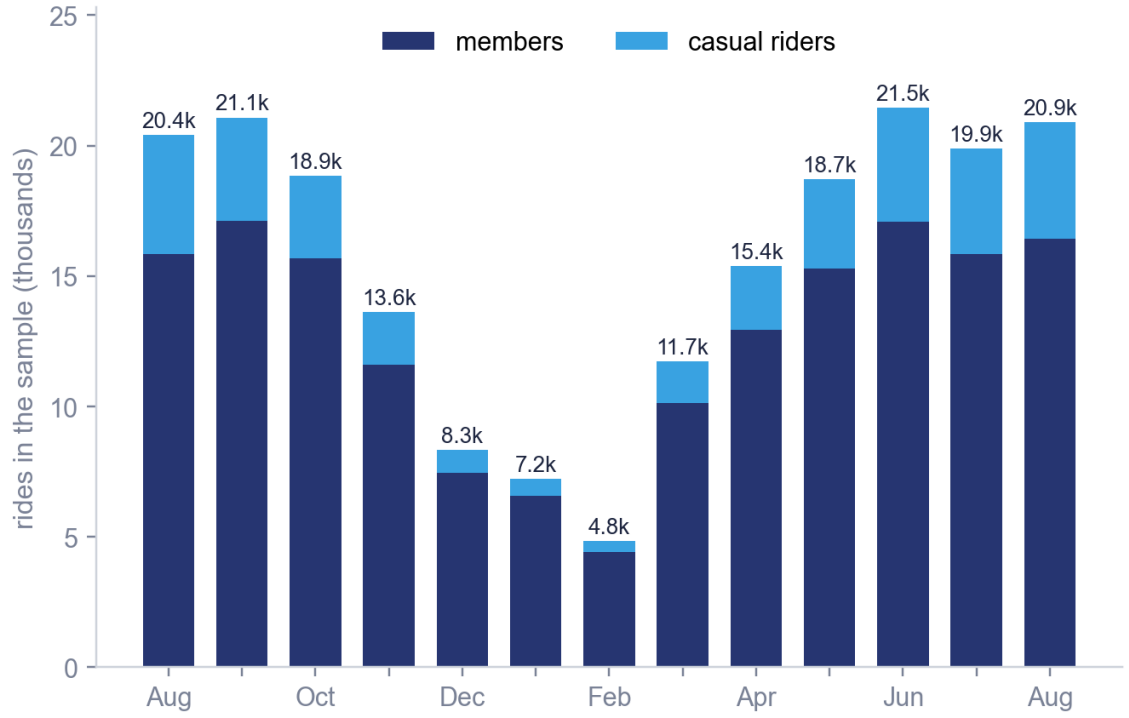
- station-days: 100,000 (demand), 150,000 (shortage models)
- benchmark and stress test: all station-days

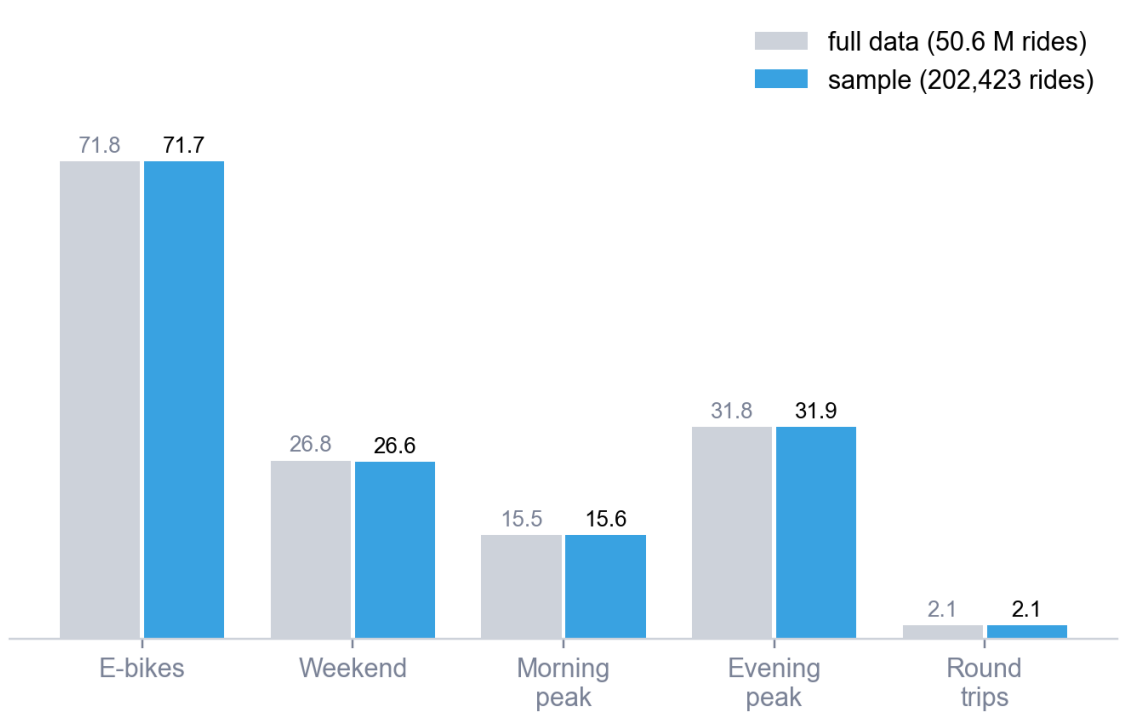
1 Sources: Citi Bike System Data (trips); E. Rosenthal, Kaggle, CC BY 4.0, from the Citi Bike GBFS feed (station status); Meteostat, Central Park (weather); Apple App Store, US (reviews).

2 pandas merge on the station name (the status data has no trip ids). Removed after the join: 9,406 incomplete station-days (< 360 snapshots) and 40,734 station-days of inactive stations (< 365 departures a year).

The 0.4% stratified sample keeps the characteristics of all rides

Rides drawn at random within each month and rider type (26 groups).[1] Variables we did not stratify on match the full data within 0.2 points.[2]





1 pandas: trips.groupby(['month', 'member_casual']).sample(frac=0.004, random_state=42) on the 50.6 M clean rides.

2 Shares in the sample and in the full data differ by less than 0.2 points. Stratifying keeps the seasonality and the 18% share of casual riders; 0.4% still leaves at least 4,800 rides per month.

02

PART 2 OF 4

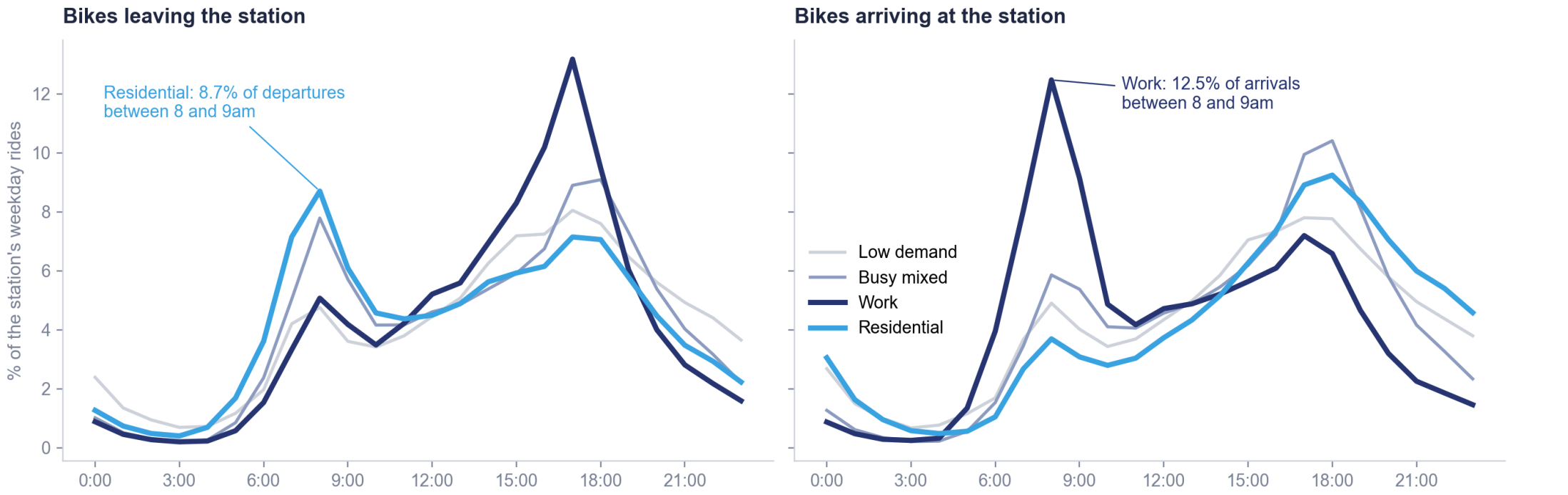
Understanding the network

Four types of stations, what drives their demand, and who rides.



Clustering the hourly profiles gives four types of stations

Average share of a station's weekday rides in each hour, for the four types found by k-means.[1]



Why four types: the statistics give no clear number (the silhouette falls from 0.15 with two clusters to 0.11 with four).[2] We chose four for business use: each type has a distinct role for the trucks.

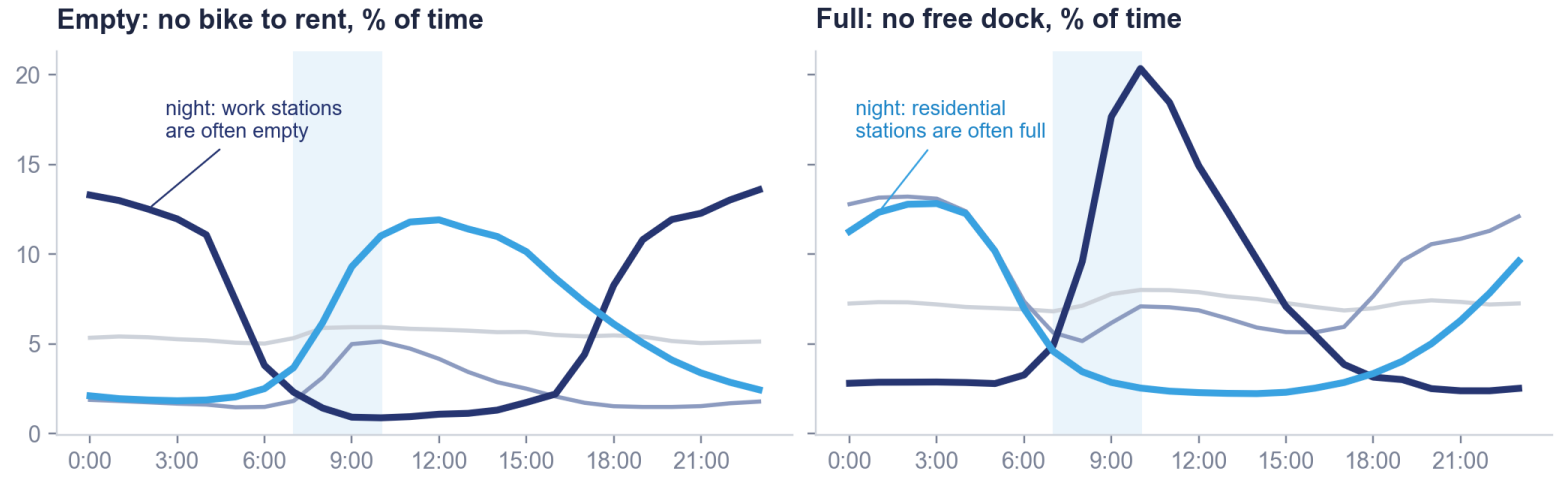
1 48 variables per station (hourly shares of departures and arrivals), StandardScaler, PCA(n_components=0.90): 27 components; KMeans(n_clusters=4, n_init=10, random_state=42).

2 Silhouette: 0.15 at k = 2, 0.13 at k = 3, 0.11 at k = 4, 0.09 at k = 5. The elbow curve has no clear break (appendix A6).

Residential stations empty in the morning; work stations fill up

The four types, where they are,[1] and how often their stations are empty or full during a weekday.[2]

Type	Main areas	Stations	Rides / day	Docks	Casual	Net flow / yr
Residential	upper Manhattan, Brooklyn, Queens	628	26	24	21%	-77
Work	Midtown, Financial District	337	79	43	20%	-10
Busy mixed	downtown, north Brooklyn	842	91	37	18%	+79
Low demand	outer areas	511	16	23	24%	-39

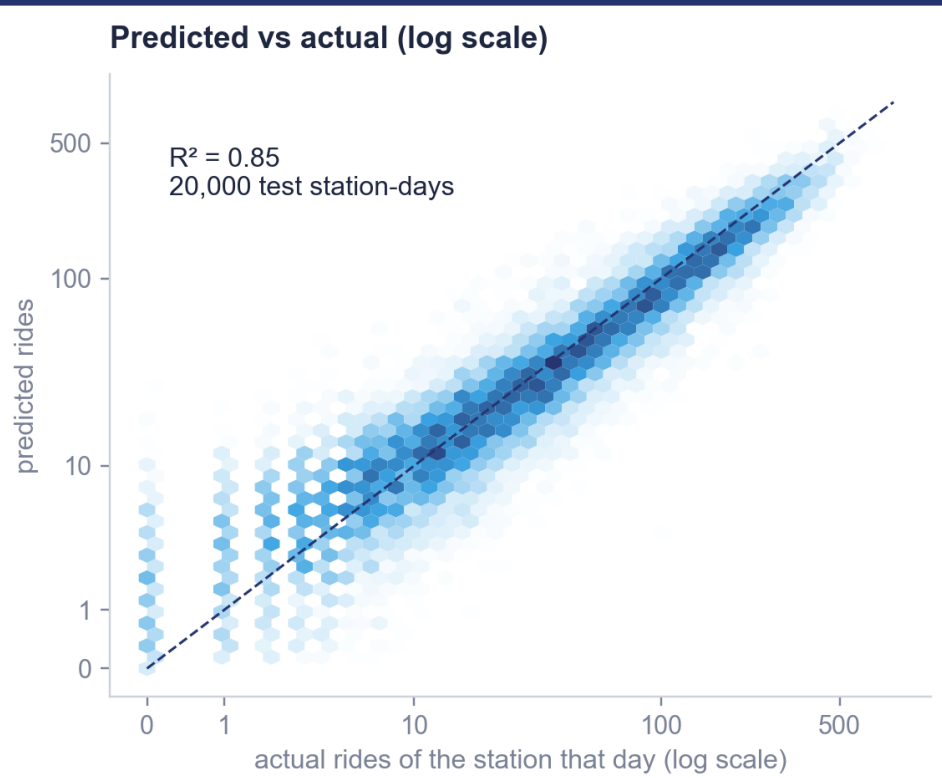


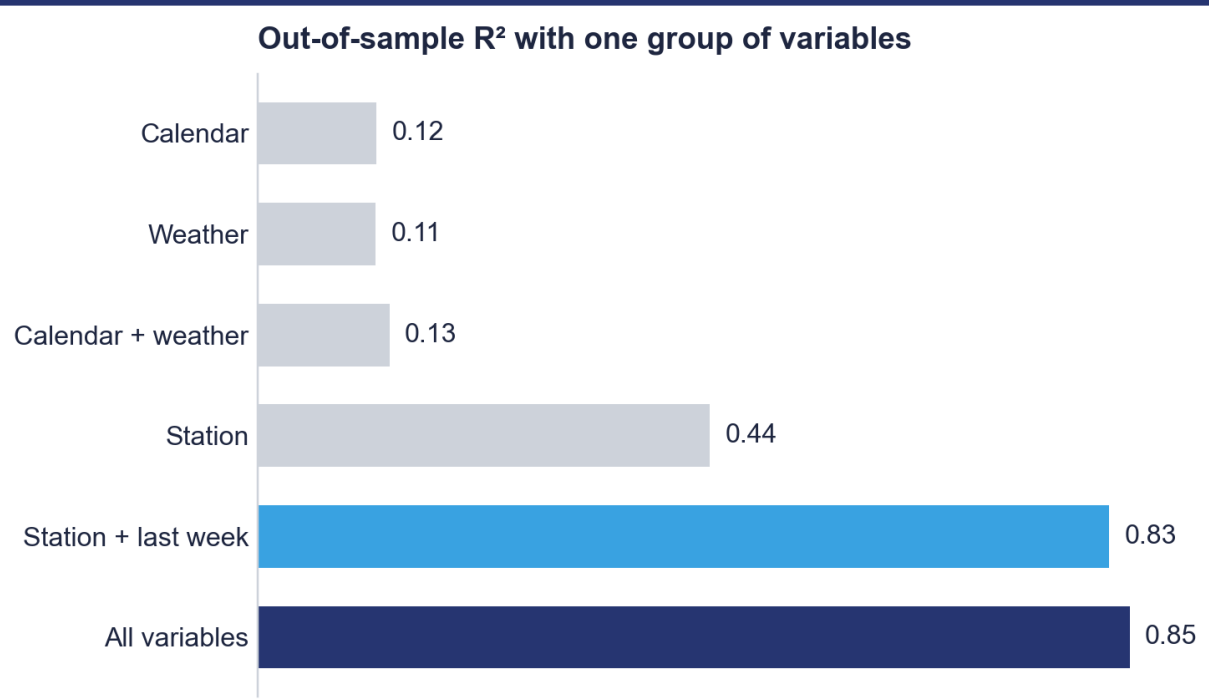


1 Averages per station; rides per day over the 396 days; net flow = rides ending − rides starting over the year. k-means only saw the hourly profiles, not the coordinates. Colours as in the table.
2 Weekday 7–10am, share of time empty: residential 6.4%, work 1.6%, busy mixed 3.3%, low demand 6%; full: residential 3.5%, work 10.4%. At night the cycle reverses.

A station's own history predicts its demand better than the calendar

Left: predicted against actual rides on test station-days.[1] Right: share of the variation explained by one group of variables at a time.[2]



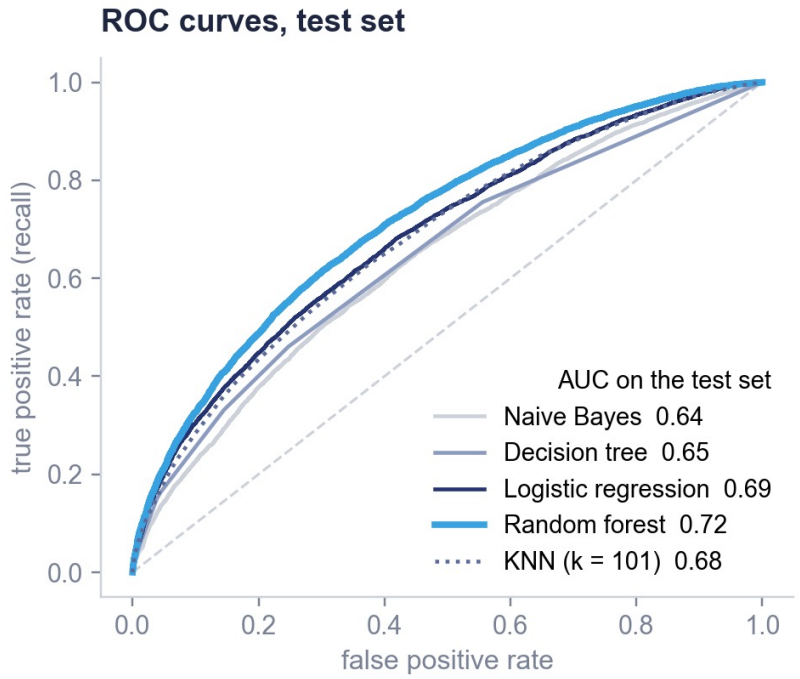


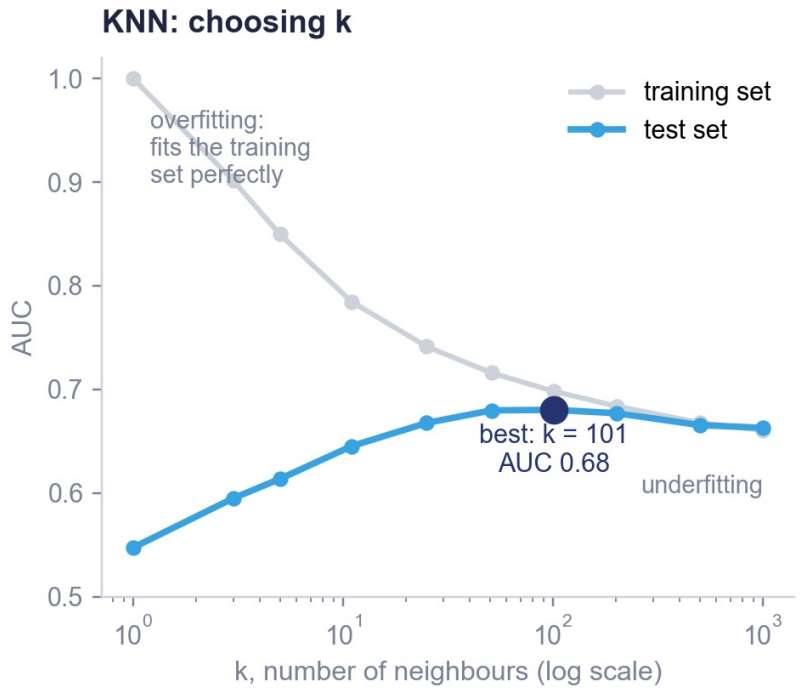
1 LinearRegression on 100,000 sampled station-days, 80/20 train_test_split (random_state = 42); R² on the test set 0.85 in logs, 0.64 without the log; 5-fold CV R² 0.850. Lasso keeps 15 of 26 variables with the same R².

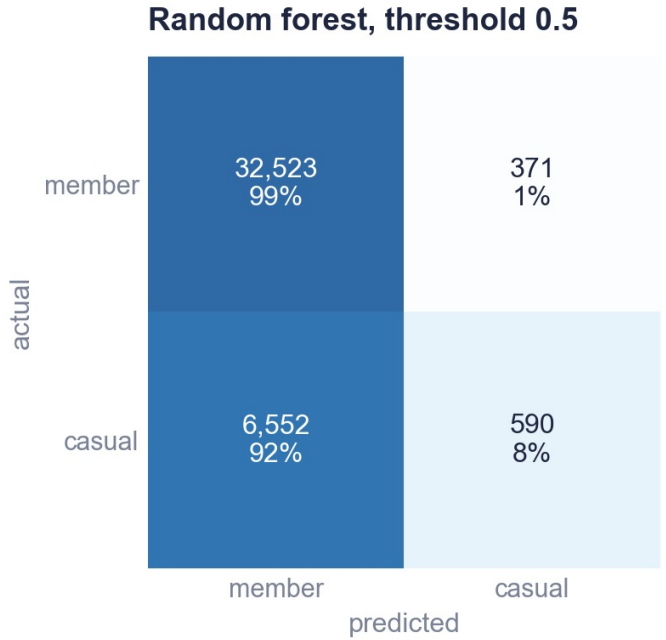
2 Calendar: day of the week, month, holiday. Weather: temperature, rain, wind. Station: capacity and type. City level (one row per day): weather alone R² = 0.78 (appendix A8).

From a single ride, members and casual riders are hard to tell apart

Each ride classified as member or casual from duration, distance, hour, weekend, bike type, weather and station type (40,036 test rides).[1]







Why KNN: it shows the bias–variance trade-off when choosing k. The random forest stays the best model.[2]

Business use: 28% of casual rides are on weekdays at commuting peaks (members: 40%): these riders are the target for a membership offer.[3]

1 AUC: probability that a random casual ride gets a higher score than a random member ride (0.5 = chance, 1 = perfect).
2 KNeighborsClassifier on standardised variables, trained on 20,000 rides; only k was varied, so this is an exploratory comparison. Random forest: 100 trees, min_samples_leaf = 20.
3 At threshold 0.5 the random forest labels 6,552 of 7,142 casual test rides as member rides. Commuting peaks: weekdays, 7–10am and 4–8pm.

03

PART 3 OF 4

Predicting tomorrow morning

Which stations will be empty tomorrow, and how to plan the trucks.



The shortage model only uses what is known the evening before

Target: the station is empty for at least one hour of tomorrow's 7–10am window.[1] It happens on 4.7% of station-days, so always answering "not empty" would be 95% accurate and useless.

Today — what the station did → **Tonight** — the model ranks every station → **Tomorrow 7–10am** — empty for at least one hour?

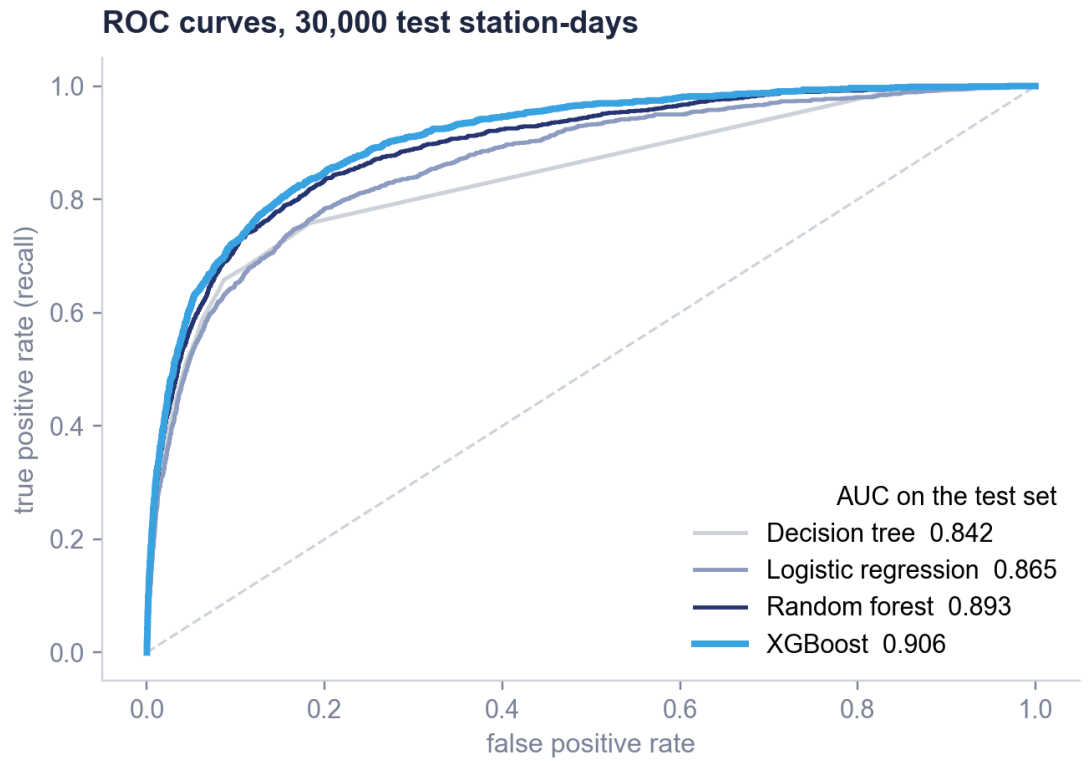
Inputs, known tonight	Variables
Station's state today	share of the morning and evening it was empty; share it was full
Today's rides	departures, arrivals, morning net flow
Station	capacity (docks), station type from the clustering
Calendar of tomorrow	day of the week, weekend, public holiday
Weather of tomorrow	average temperature, rain[2]

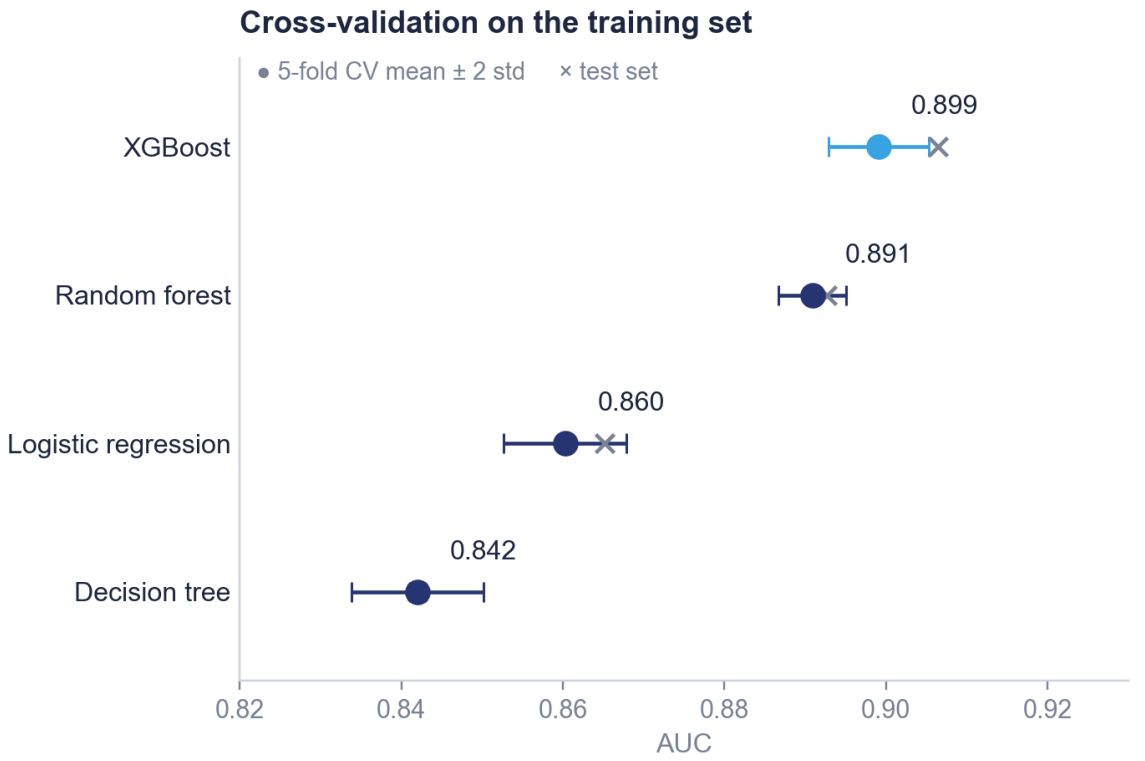
Design of the test	Size
Station-days sampled[3]	**150,000**
Training set (5-fold cross-validation)	**120,000**
Test set, never used for training	**30,000**
Empty tomorrow in the test set	**4.6%**
Models compared · main metric	**4 · AUC**

1 Empty ≥ 1 hour = empty in at least 33% of the 7–10am snapshots. A station empty for more than two thirds of this morning has a 42% chance of this tomorrow.
2 Observed weather is used in place of tomorrow's forecast. Removing the weather variables leaves the AUC essentially unchanged.
3 data.sample(150000, random_state=42), then train_test_split(test_size=0.2, random_state=42).

XGBoost gives the best ranking of tomorrow's empty stations

Left: ROC curves of four models on the 30,000 test station-days.[1] Right: AUC in 5-fold cross-validation on the training set.[2]



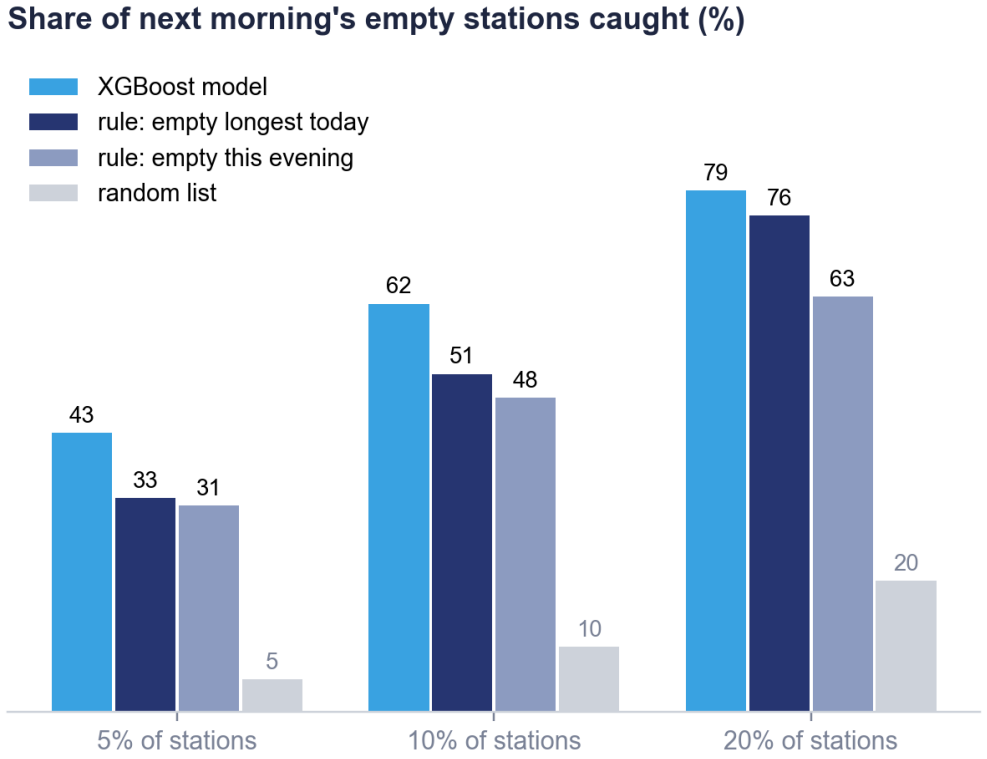


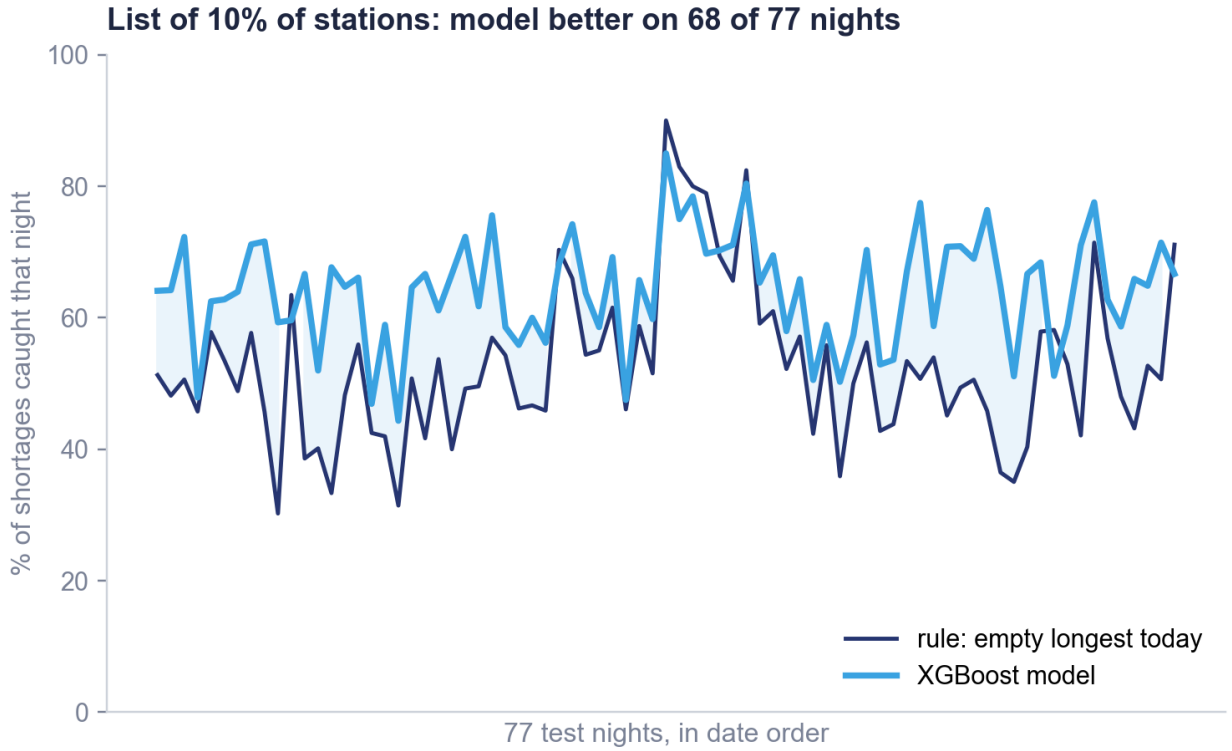
1 AUC: how often the model ranks a station that will be empty above one that will not. XGBoost 0.906; five other random splits give 0.896–0.903.

2 DecisionTreeClassifier(max_depth=5); LogisticRegression; RandomForestClassifier(100 trees, min_samples_leaf=20); XGBoost (max_depth=4, eta=0.3, 100 rounds). Cross-validation: KFold(5, shuffle=True, random_state=10).

Night by night, the model finds more shortages than a simple rule

Without a model, an operator would send the trucks to the stations empty longest today.[1] Model and rule get the same number of stops on 77 nights the model never saw.[2]



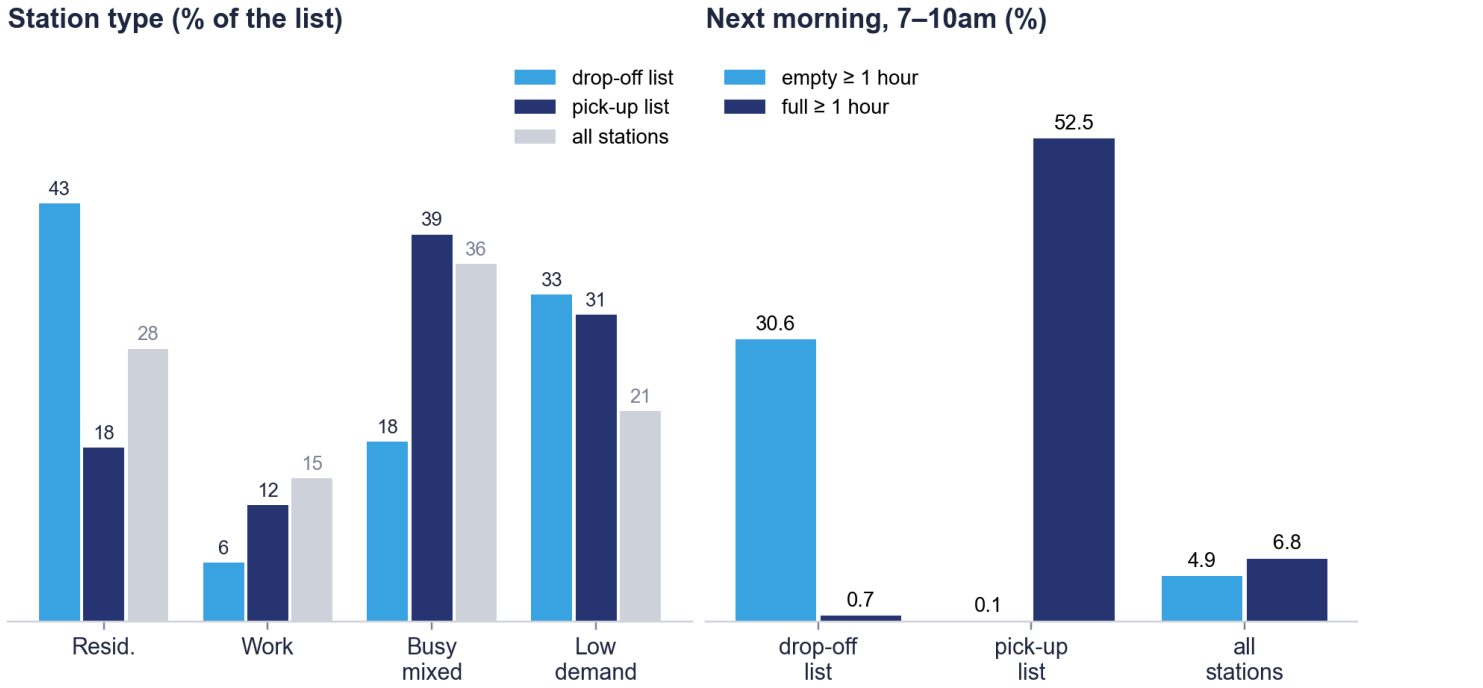


1 With 10% of stations: model 62% of shortages (hit rate 31%), rule 51% (25%), random list 10% (4.9%). AUC on the test nights: model 0.905, rule 0.855. Second rule, "empty longest this evening": 48%. Hit rate = listed stations really empty the next morning.

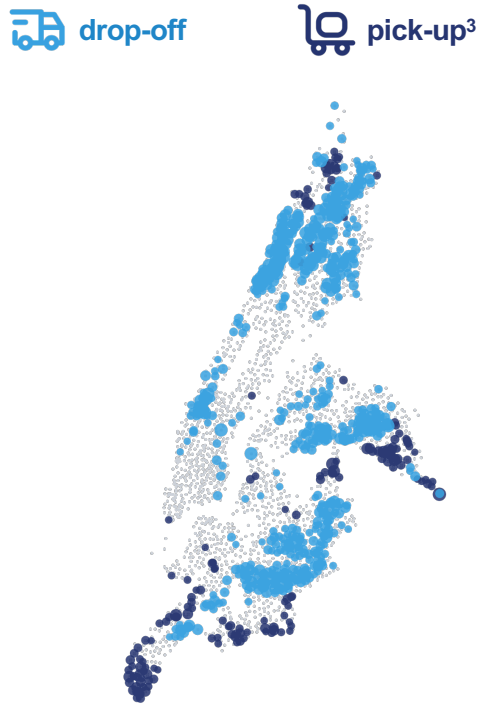
2 Split by date: 80% of the nights to train, 20% (77 nights) to test. Each night every method lists the same share of stations (5%, 10% or 20%).

The same scores give the trucks a drop-off list and a pick-up list

Drop-off: the 10% of stations at highest risk each night. Pick-up: stations full this evening that the model ranks at low risk.[1] Test nights.



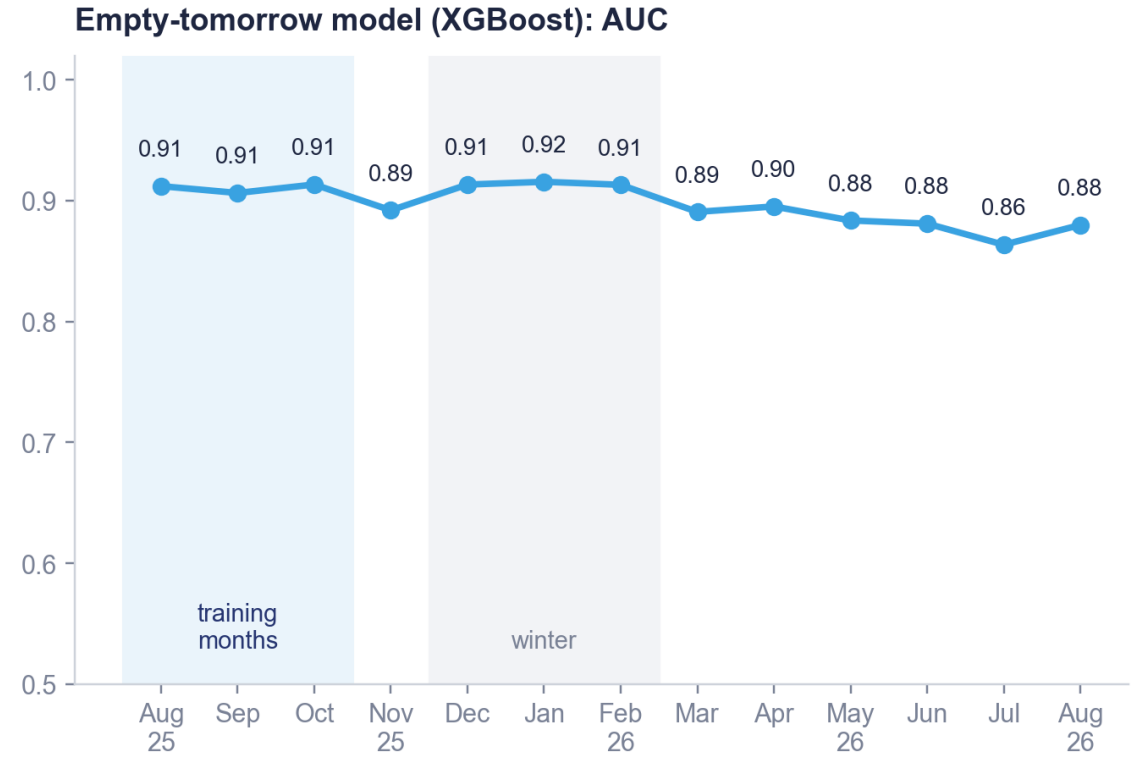
Pick-up stations the next morning: 0.1% empty, 53% full again.[2]

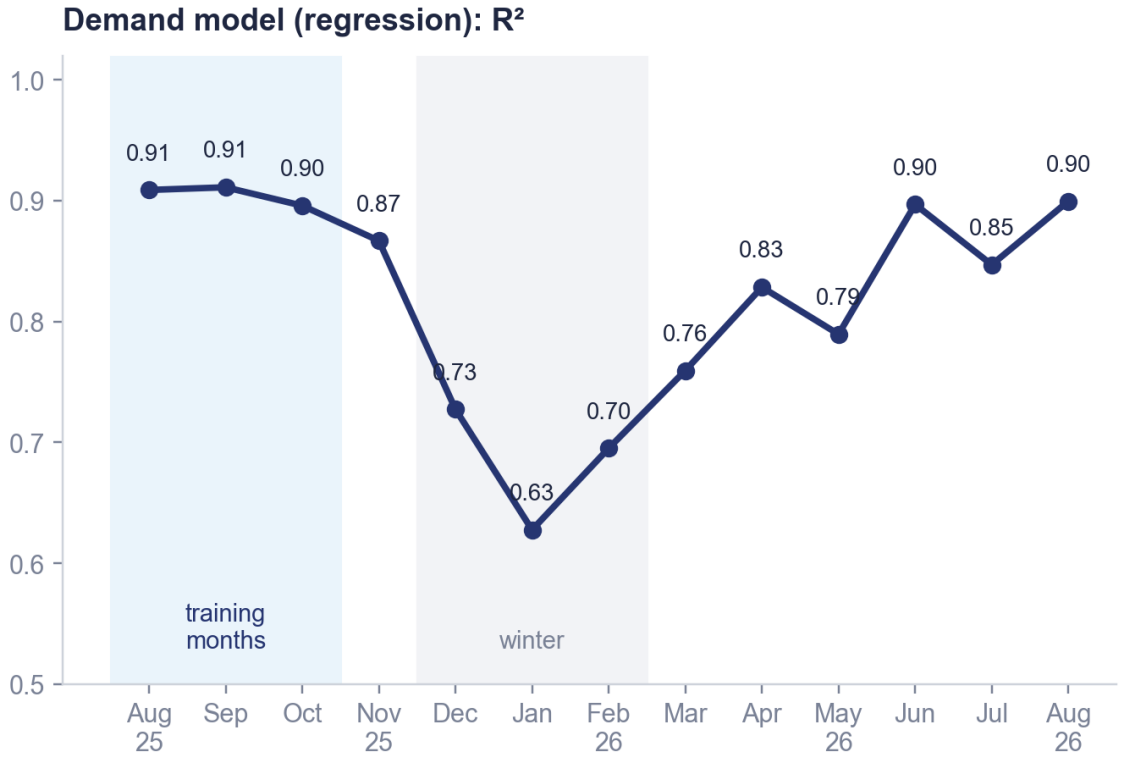


1 Full this evening: full for at least one third of 4–8pm. Pick-up list: the half of these stations with the lowest predicted risk, about 67 stations a night.
2 Of all stations full this evening, 1.0% are empty and 39% full the next morning. Before collecting, the driver checks the live availability and leaves a reserve.
3 Map: stations on the drop-off list (blue) or on the pick-up list (navy) on at least 15% of the test nights.

Trained on autumn data, the shortage model keeps its accuracy all year

Both models trained on August–October 2025 only,[1] then tested month by month on the following year.[2]





1 Time-based split on all station-days: trained on Aug–Oct 2025, tested on each later month. Same XGBoost and linear regression settings as before.

2 AUC by period: Aug–Oct 0.911, Dec–Feb 0.914, Jun–Aug 2026 0.877. Demand R²: 0.905, 0.686, 0.881.

04

PART 4 OF 4

What Citi Bike should do

Results, recommendations and limits.



Summary of results

What the analysis found, and what it means for Citi Bike's trucks and customers.

WHAT WE FOUND	WHAT IT MEANS FOR CITI BIKE
Shortages can be predicted the evening before With data known at night, XGBoost reaches an AUC of 0.91[1] on unseen days. A list of 10% of stations contains 62% of next-morning shortages; the simple rule's list 51%.	Rank all stations every evening and plan the night route from the top of the list.
Bikes are missing mostly at residential stations Residential stations are 28% of station-days but 43% of the drop-off list; low-demand stations another 33%.	Refill residential stations first, then the other high-risk stations, including low-demand ones.
Bikes can be collected at stations full tonight Of the stations full this evening and ranked at low risk, 0.1% run empty the next morning and 53% are full again.	Collect there, after checking live availability and leaving a reserve; it also frees docks.
Shortages persist; weather sets the city's volume Empty for more than two thirds of this morning: 42% chance of a shortage tomorrow. 10% of stations = 47% of empty time. Weather: 78% of the city's daily rides.	Fix chronic stations structurally; use the weather forecast to anticipate city-wide demand.
Members and casual riders overlap Best AUC 0.72 (random forest). 28% of casual rides are on weekdays at commuting peaks, against 40% of member rides.	Offer memberships to casual riders who repeatedly ride at commuting peaks.

1 AUC: probability that the model ranks a station that will be empty above one that will not (0.5 = random, 1 = perfect). Simple rule: visit the stations that were empty longest today.

Tonight's trucks should start from residential stations

Our answer to the business question: commuter (residential) stations first, refilled with bikes from stations that are full tonight and at low risk.

01

Predict

Run the model every evening

Rank all stations by the probability of being empty tomorrow and plan the night's route from the top of the list.

62%
of shortages caught with 10% of stations (simple rule: 51%)

02

Drop off

Residential stations first

Refill the residential stations at risk before 7am, then the other high-risk stations, including low-demand ones. Extra capacity on Mondays and in winter.

43%
of the drop-off list is residential, against 28% of station-days

03

Collect

Full, low-risk stations

Collect from stations full this evening that the model ranks at low risk, after checking live availability and leaving a reserve.

0.1%
run empty the next morning; 53% are full again

04

Fix

Chronic stations for good

Where a station is empty almost every day, add docks or bikes, or a peak-hour valet, instead of a daily truck.

47%
of empty morning time is at 10% of the stations

Also: use weather forecasts to anticipate city-wide demand; explore membership offers for casual users who repeatedly ride during weekday commuting peaks (28% of casual rides).

Limits of the analysis and next steps

What our data cannot tell, and what we would do before a roll-out.

Limits

No truck data

We cannot compare our lists with Citi Bike's actual dispatch, and today's status already includes today's rebalancing.

Observed weather, not the forecast

Removing weather leaves the shortage model's AUC essentially unchanged.

Daily, not hourly

"Empty for one hour of 7–10am" is a threshold we chose.

Small text sample

500 iOS reviews, mostly negative; stations matched by name (99.2% of rides).

Next steps

Add the rebalancing data

Measure what the trucks already do, and the gain of our lists over it.

Predict hour by hour

Plan daytime truck rounds too, not only the night.

Pilot in one borough

Compare shortages with and without the lists for a few weeks.

Retrain every month

The AUC drops slightly a year later as the network changes.

Tonight: refill residential stations, with bikes from stations that are full and at low risk.

A model using only what is known in the evening ranks tomorrow's empty stations well (AUC 0.91), also in winter. With 10% of stations it finds 62% of them, against 51% for the simple rule.

Thank you. Questions?





Residential stations, where tonight's trucks should go first

A

Appendix

A1 Our five hypotheses against the results

A2 Cleaning rules

A3 Rides over the year, and where they start

A4 Members and casual riders compared

A5 When people ride

A6 Choice of k: PCA, elbow, silhouette

A7 Where and when shortages happen

A8 Weather and demand

A9 Regression coefficients and Lasso

A10 Riders: importance and resampling

A11 Persistence and the target

A12 Decision threshold

A13 What drives the prediction

A14 Cumulative gains on the random test set

A15 Topic model of the reviews (LDA)

Our five hypotheses against the results

What we expected before the analysis, what we found, and where the evidence is.

	HYPOTHESIS	RESULT	EVIDENCE	SLIDES[1]
H1	**Time explains a station's demand better than the station**	**Not at station level**	Station-day R^2: calendar 0.12, station 0.44, station with last week's rides 0.83. For the whole city, weather explains 78% of daily rides.	11, A8
H2	**Trip patterns separate members and casual riders**	**Partly**	Best AUC 0.72 (random forest), KNN 0.68 at its best k. 28% of casual rides are at commuting peaks, 40% of member rides.	12, A4
H3	**Stations fall into a few behavioural types**	**Supported**	Four types with distinct profiles and opposite daily cycles. Low silhouette (0.11 at k = 4): four chosen for business use.	9–10, A6
H4	**Weather is a secondary but real driver**	**Partly**	78% of the city's daily rides, +3.5% per °C; removing weather leaves the shortage model's AUC essentially unchanged.	11, A8, A13
H5	**Models lose accuracy in another season**	**For demand only**	Demand R^2 falls to 0.63 in January; the shortage model's AUC stays around 0.91 in winter.	18

1 Slide numbers refer to this deck; codes starting with A to the appendix.

Cleaning rules and rows removed

One rule for each problem found while exploring the data, applied to every row.[1]

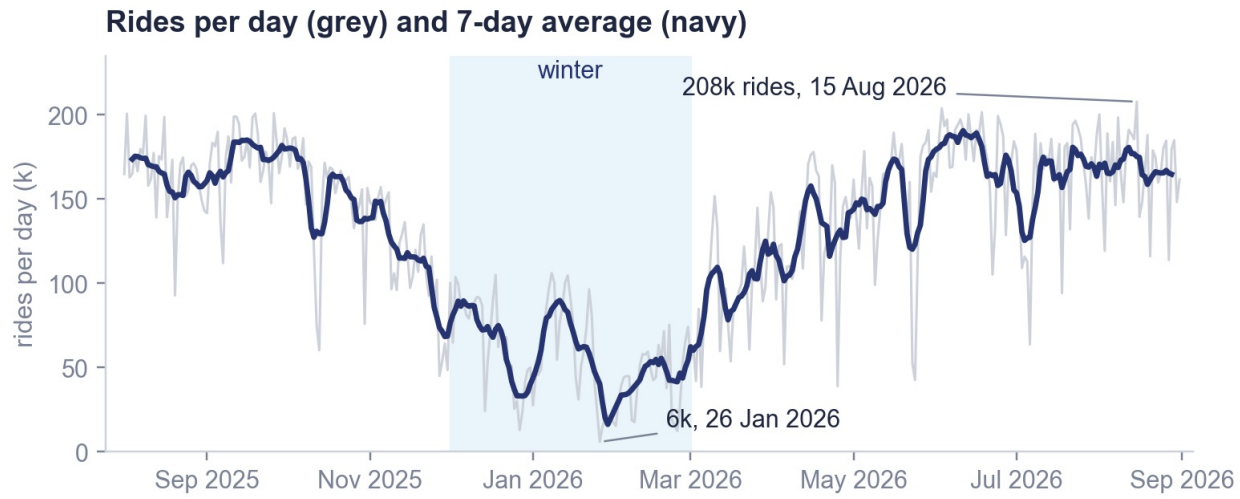
Rule	Rows removed
Trip data: 50,810,230 rides	
No start or end station	168,313
Longer than 3 hours	34,578
Duplicated ride id	512
Zero or negative duration	482
Outside Aug 2025 – Aug 2026	290
Under 1 minute, wrong coordinates	8
Removed in total	**204,183 (0.4%)**

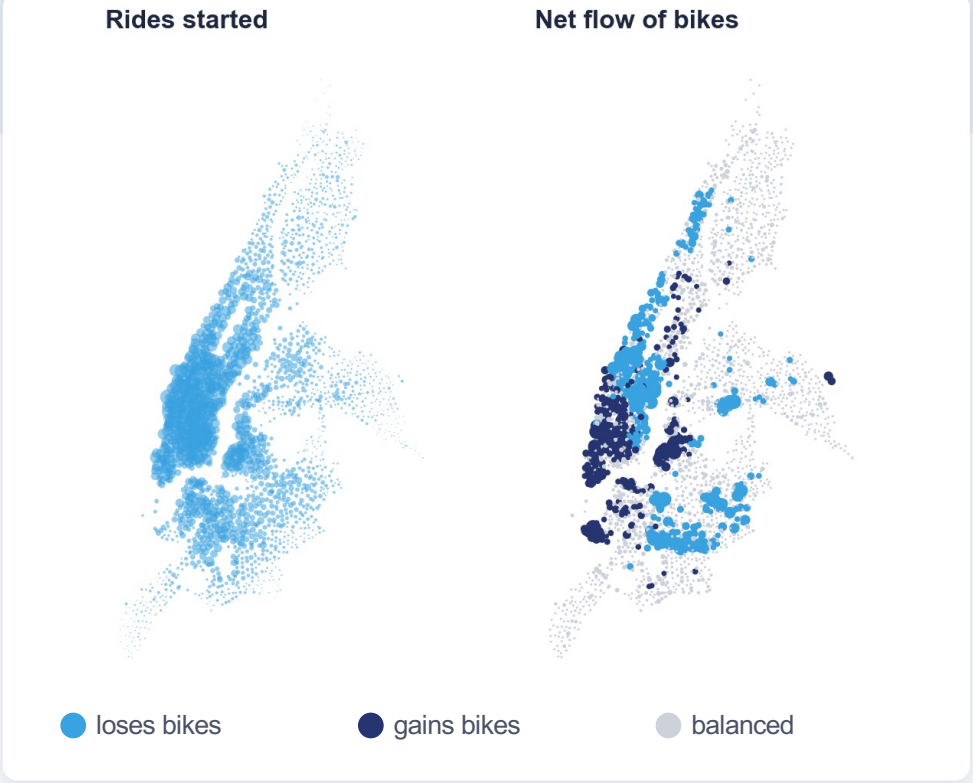
Rule	Rows removed
Station status: 1,062,555,330 snapshots	
Outside the period	389.7 M
Station not in service	19.5 M
Zero or missing capacity	8.0 M
Duplicated snapshot	2.4 M
Kept	**642.9 M**
Station-days after joining: 901,753	
Incomplete day (< 360 snapshots)	9,406
Inactive station (< 365 departures)	40,734
Kept	**851,613**

1 Details in the data cleaning report. Rules applied with pandas (dropna, drop_duplicates, boolean filters) before any aggregation.

Rides follow the seasons, and bikes drift towards the centre

Left: rides per day and share of casual riders by month. Right: rides started per station[2] and net flow of bikes over the year.[1]



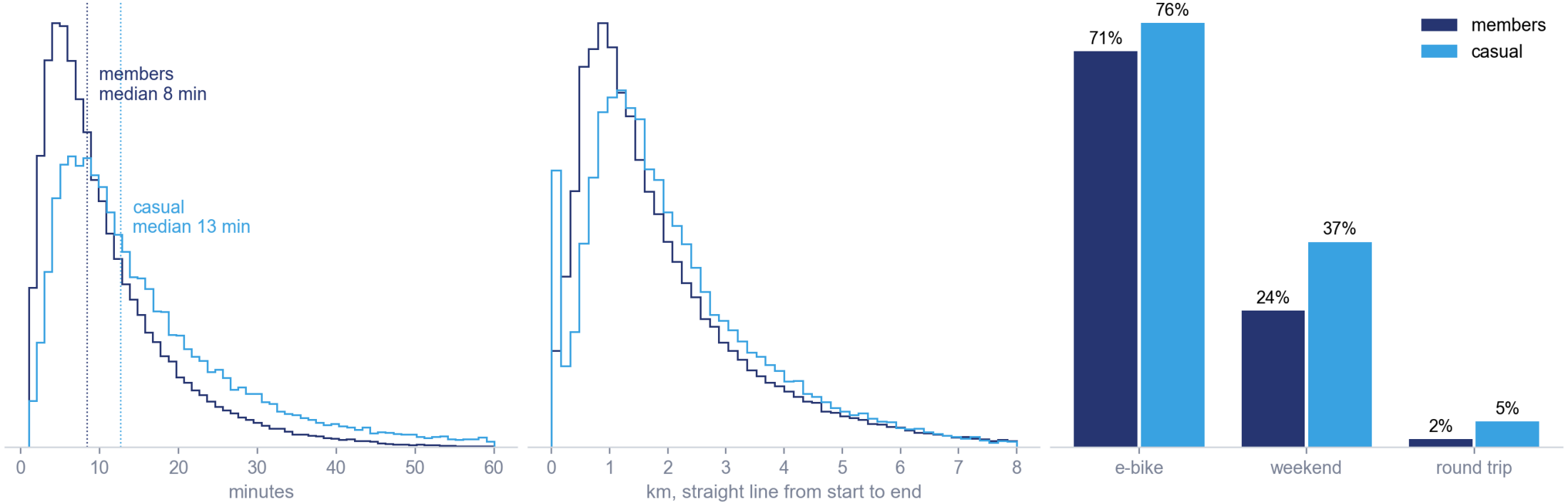


1 Net flow = rides ending − rides starting at the station over the year. Blue: loses more than 300 bikes (277 stations); navy: gains more than 300 (265); grey: in between.

2 The busiest 10% of stations account for 39% of rides. Casual share by month from the 0.4% stratified sample.

Casual rides are longer and more often at weekends, but most are short

Members and casual riders compared on duration, distance and three ride characteristics.[1] Commuting peaks: casual 28%, members 40%.[2]

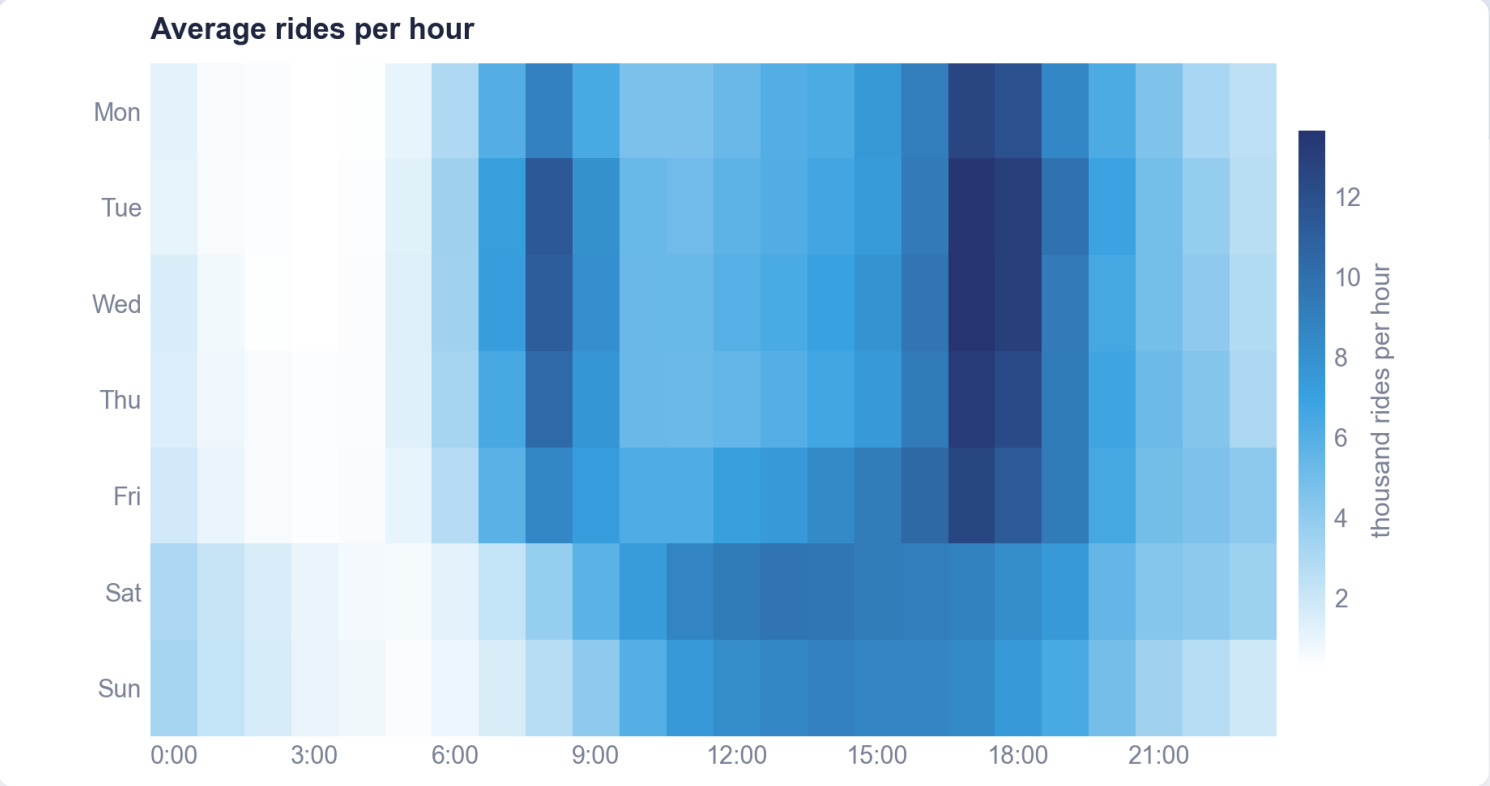


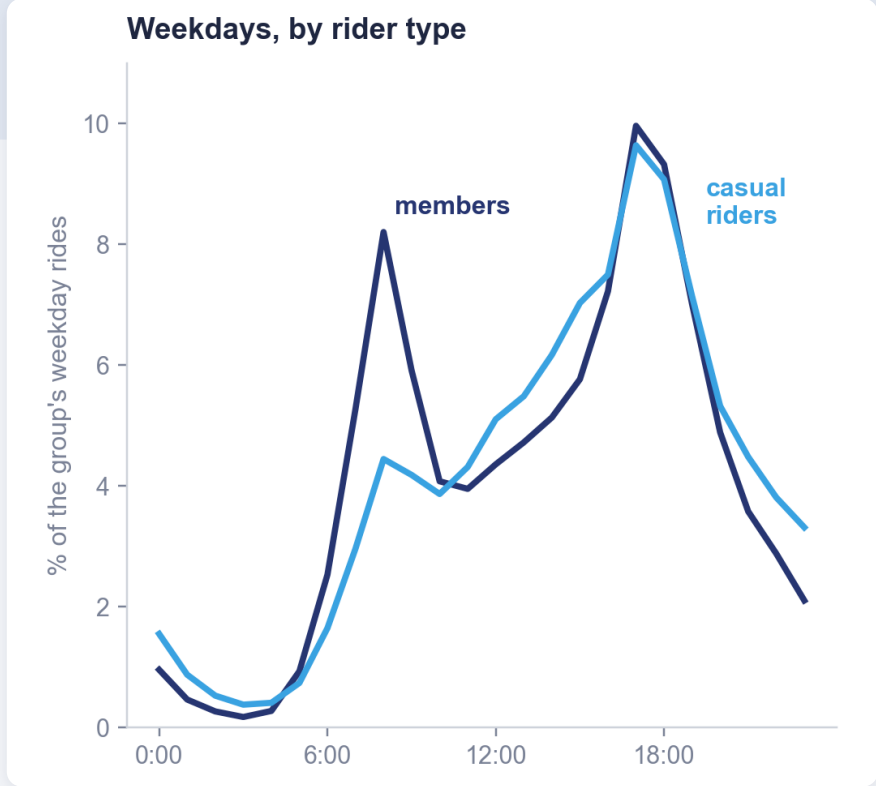
1 0.4% stratified sample (202,423 rides). Histograms normalised within each group; rides over 60 minutes and 8 km not shown. Round trip: same start and end station. Dotted lines: medians.

2 Commuting peaks: weekdays 7–10am and 4–8pm. Casual rides 28%, member rides 40%.

Weekday rides peak at 8am and 5–6pm; weekends have one midday peak

Left: average rides per hour, by day of the week and hour. Right: share of each group's weekday rides by hour.

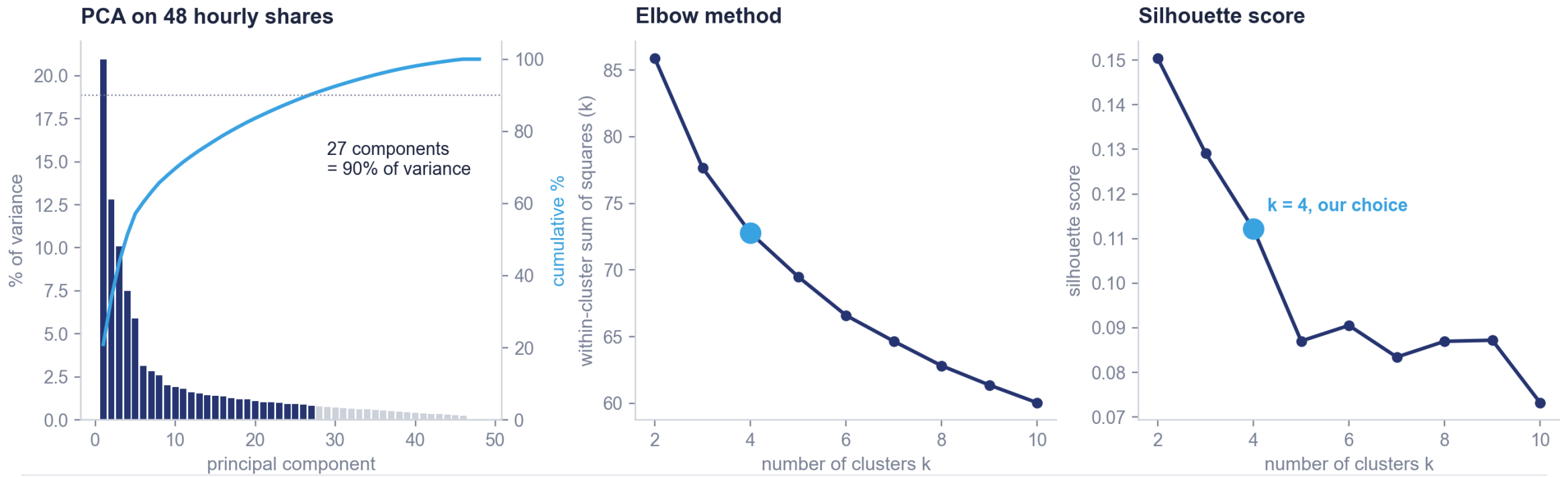




1 Average rides per hour = sample rides × 250 ÷ number of such days in the year (0.4% stratified sample).

Choice of k: the elbow and the silhouette give no clear answer

Each station is described by 48 hourly shares, standardised, reduced with PCA, then grouped with k-means for k from 2 to 10.[1]

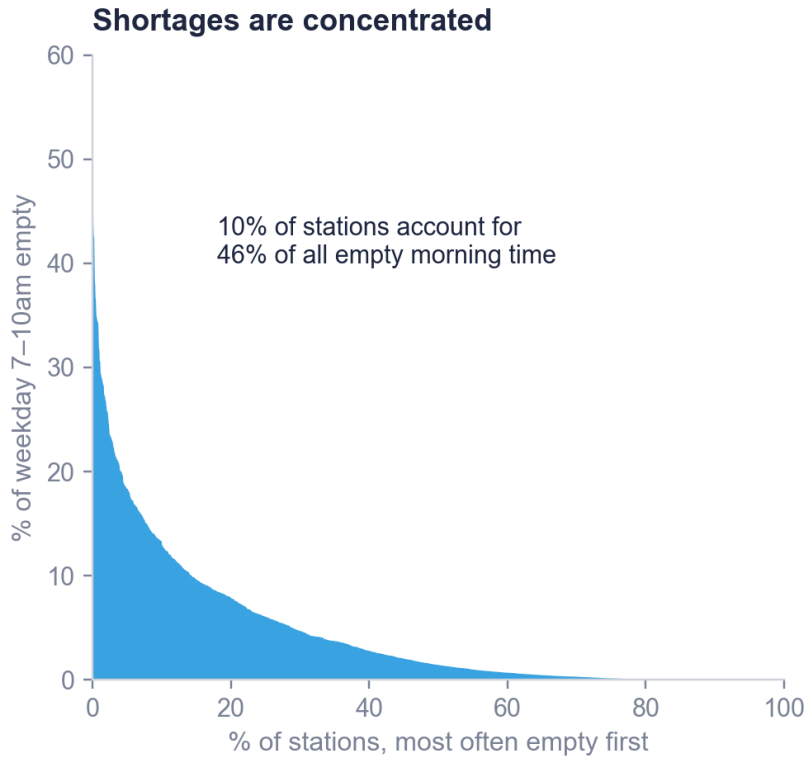


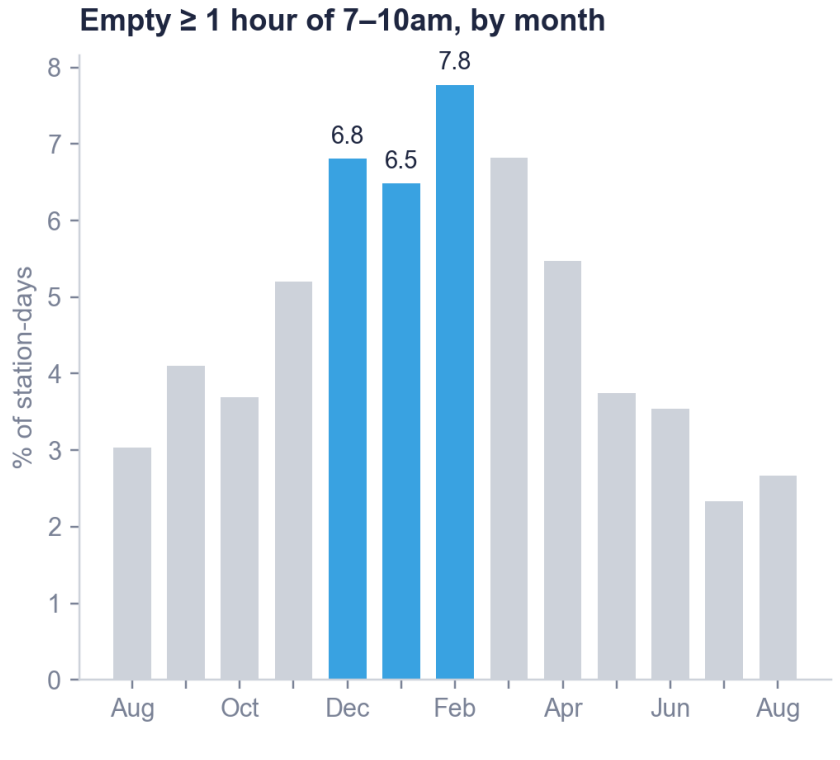
The silhouette falls from 0.15 (k = 2) to 0.11 (k = 4): we chose k = 4 for business use, since four types can be named and acted upon.

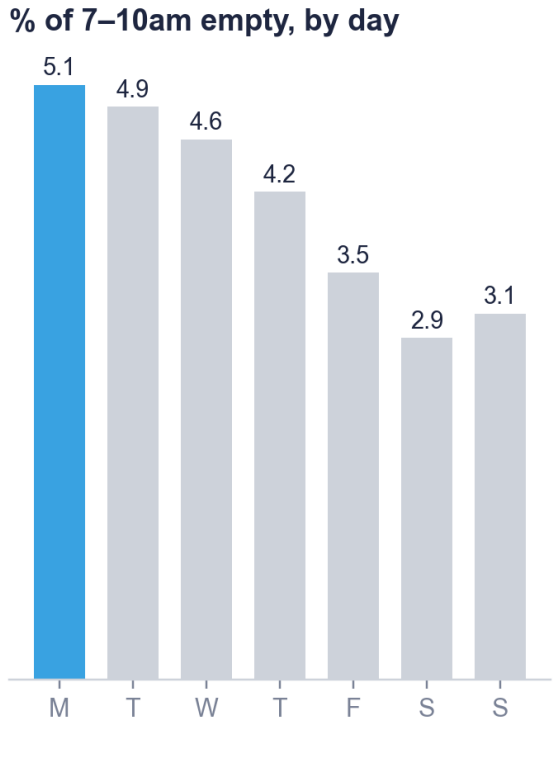
1 KMeans(random_state=42, n_init=10) on the 27 principal components. Silhouette: k = 2: 0.15, k = 3: 0.13, k = 4: 0.11, k = 5: 0.09.

Shortages are concentrated in a few stations, in winter and on Mondays

Share of the weekday morning each station is empty; station-days empty ≥ 1 hour of 7–10am by month; morning empty share by day.[1]



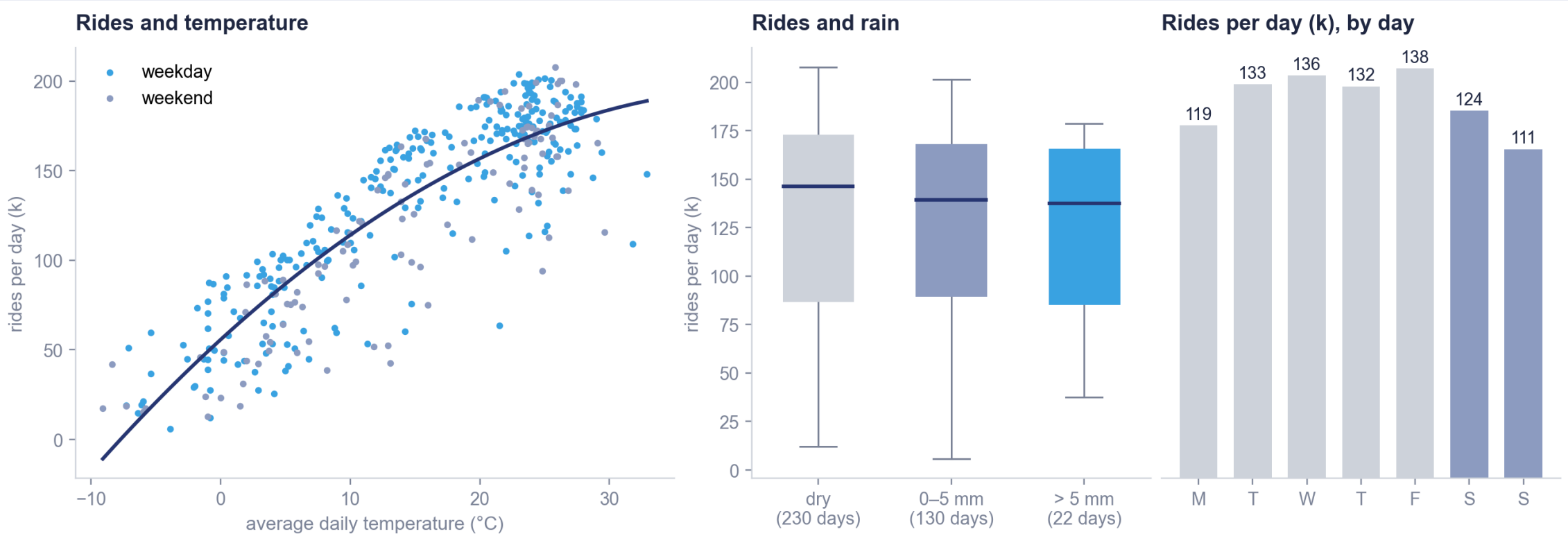




1 Empty ≥ 1 hour = empty in at least 33% of the 7–10am snapshots: 4.7% of all 851,613 station-days. This is the target of the prediction model.

Rides rise with temperature and fall with rain

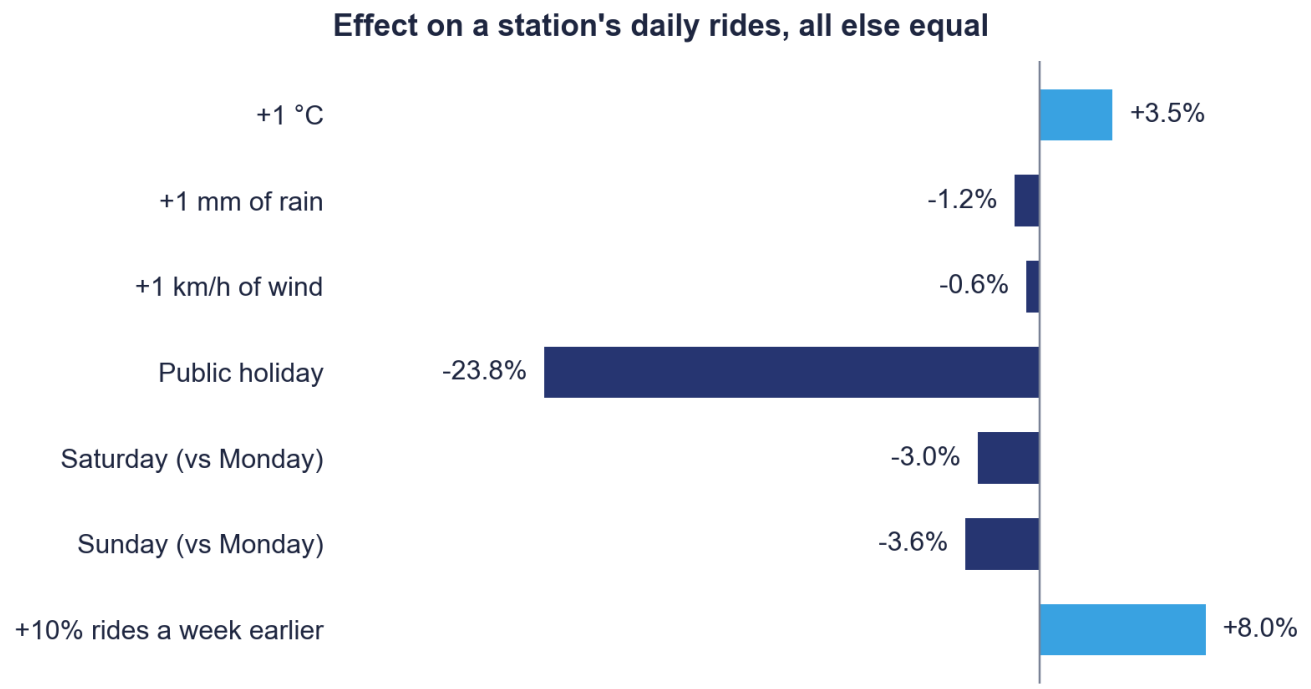
Left: one dot per day, rides against temperature. Middle: rides per day by rain. Right: average rides per day by day of the week.[1]

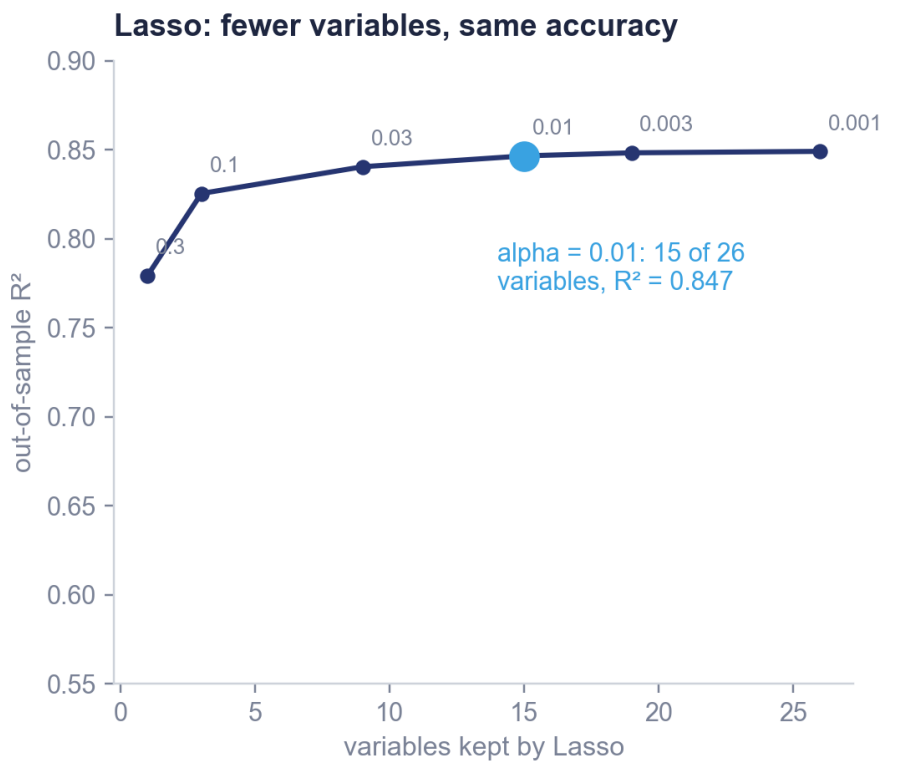


1 City-level regression of log daily rides (one row per day), test set: weather only $R^2 = 0.775$, day of the week only $R^2 = 0.007$. Navy line: quadratic fit, for illustration.

Each degree adds about 3.5% of rides; rain and holidays reduce them

Left: effect of each variable on a station's daily rides in the full regression.[1] Right: Lasso with an increasing penalty.[2]

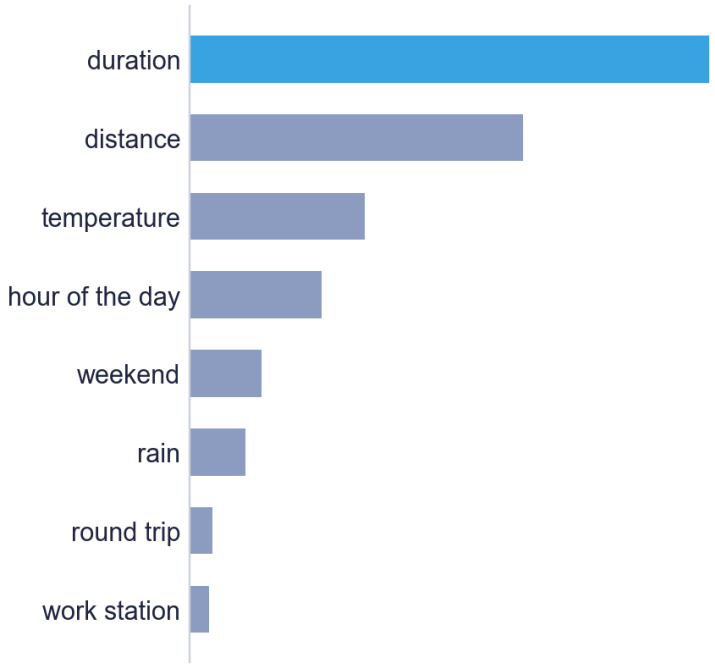


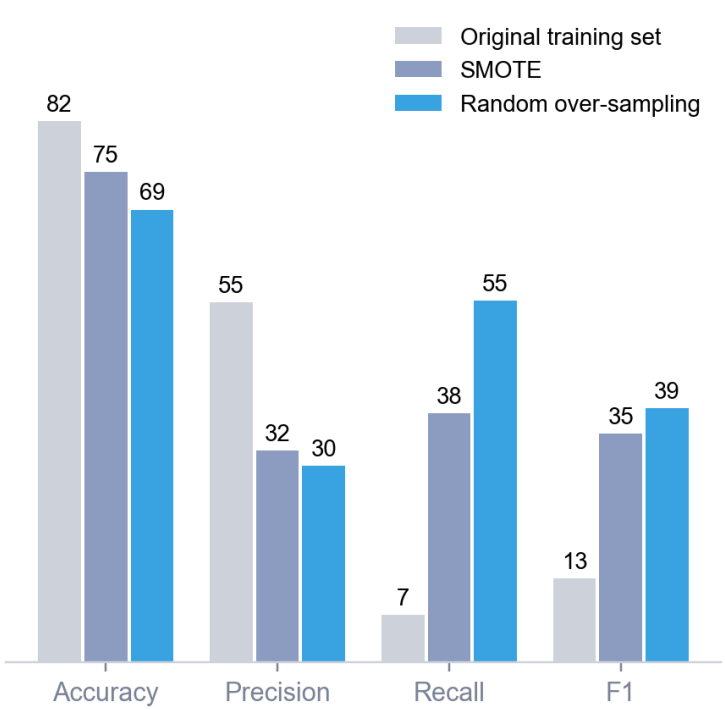


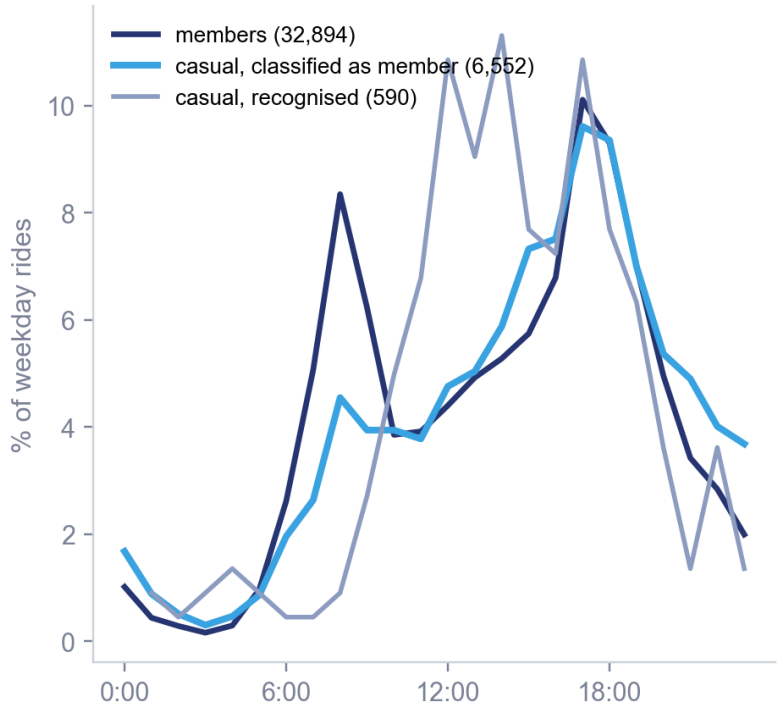
1 % effect = exp(coefficient) − 1, from LinearRegression on log(1 + departures).
2 Lasso on standardised variables; labels on the curve: alpha. Stepwise selection (mlxtend, forward and backward, 5 variables): temperature, rain, capacity, last week's rides, March.

Resampling finds more casual riders, at the cost of precision

Left: importance. Middle: logistic regression on the original, SMOTE and over-sampled training sets.[1] Right: weekday hours of test rides.[2]



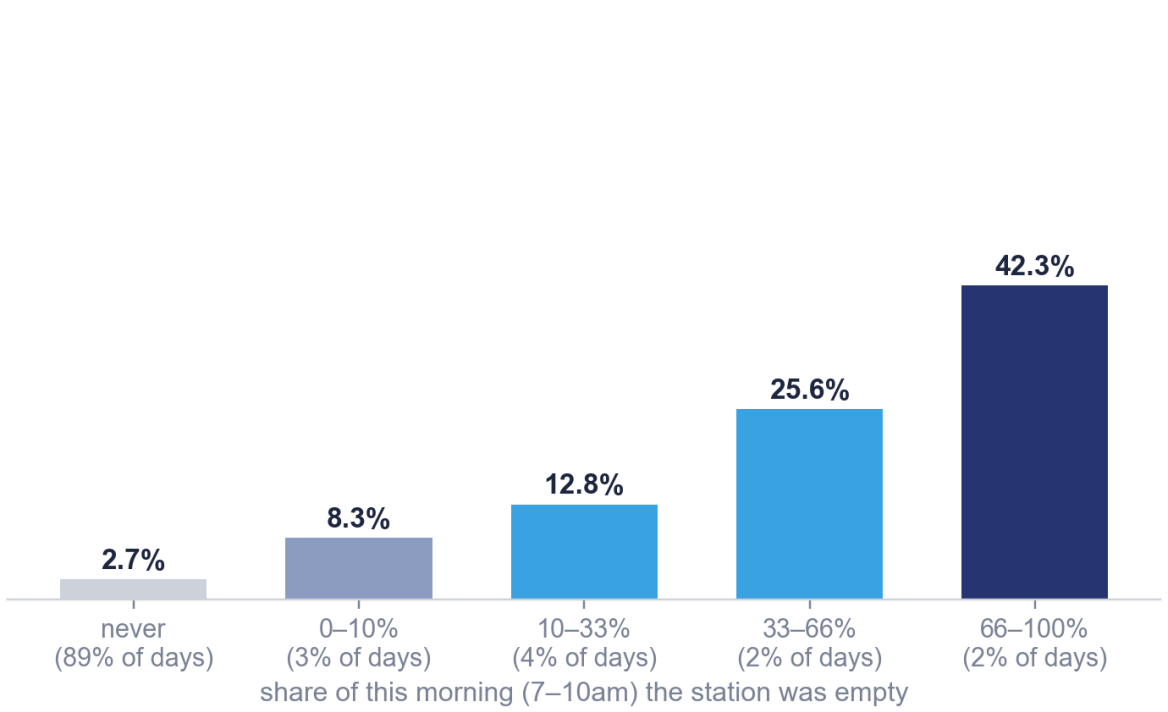


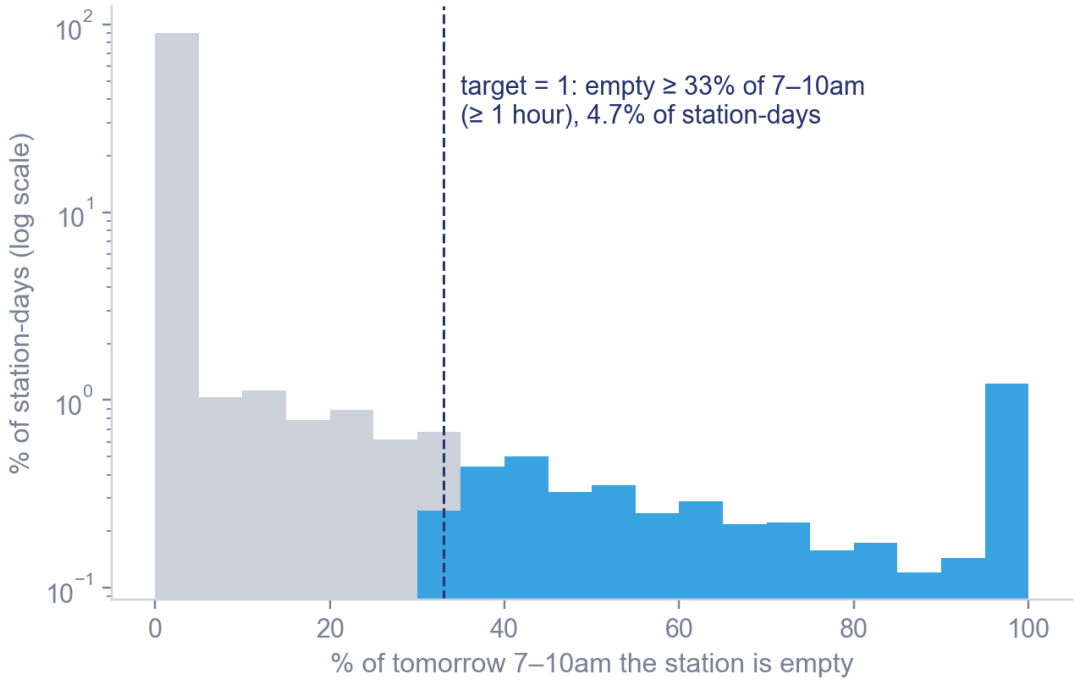


1 SMOTE and RandomOverSampler (imblearn) applied to the training set only; the test set keeps the real 18% share of casual rides.
2 Groups from the random forest at threshold 0.5.

A station empty this morning is often empty again tomorrow

Left: chance of being empty ≥ 1 hour tomorrow, by how long the station was empty this morning. Right: how the target is defined.[1]

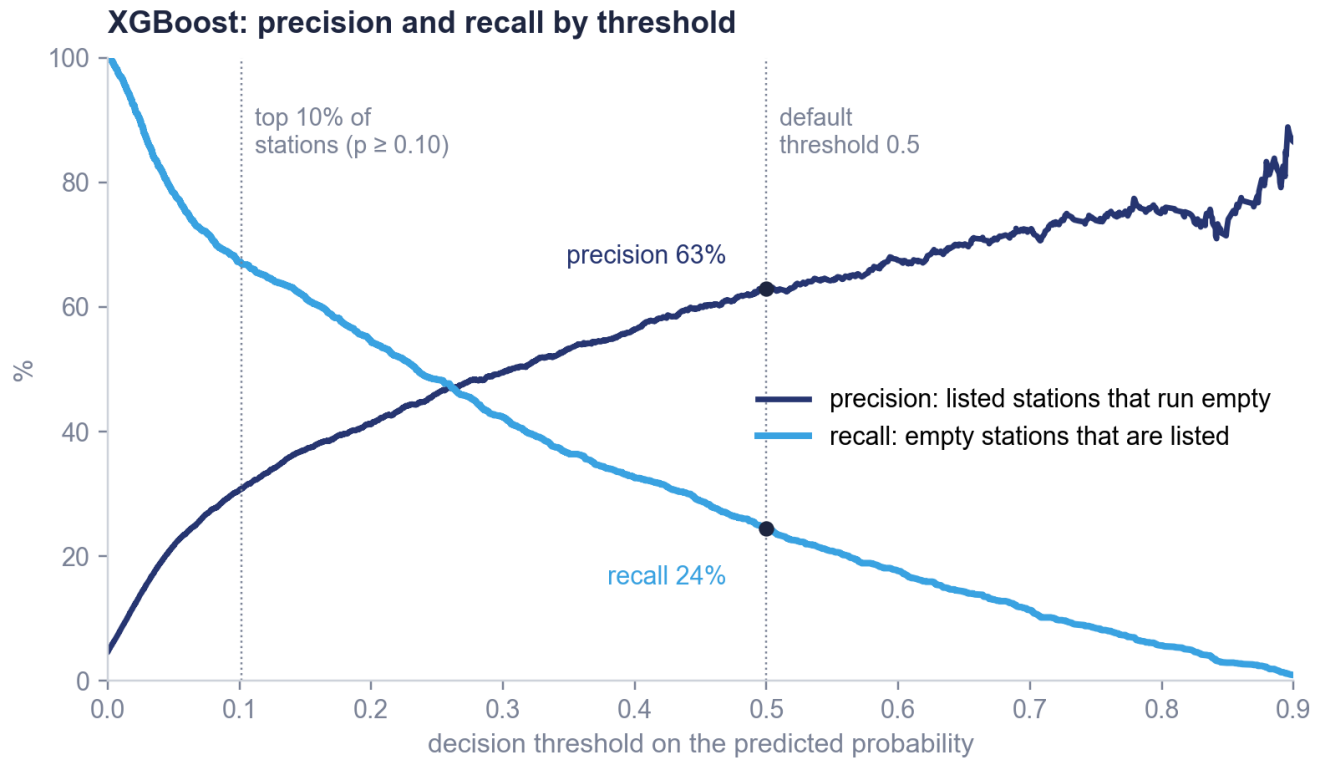


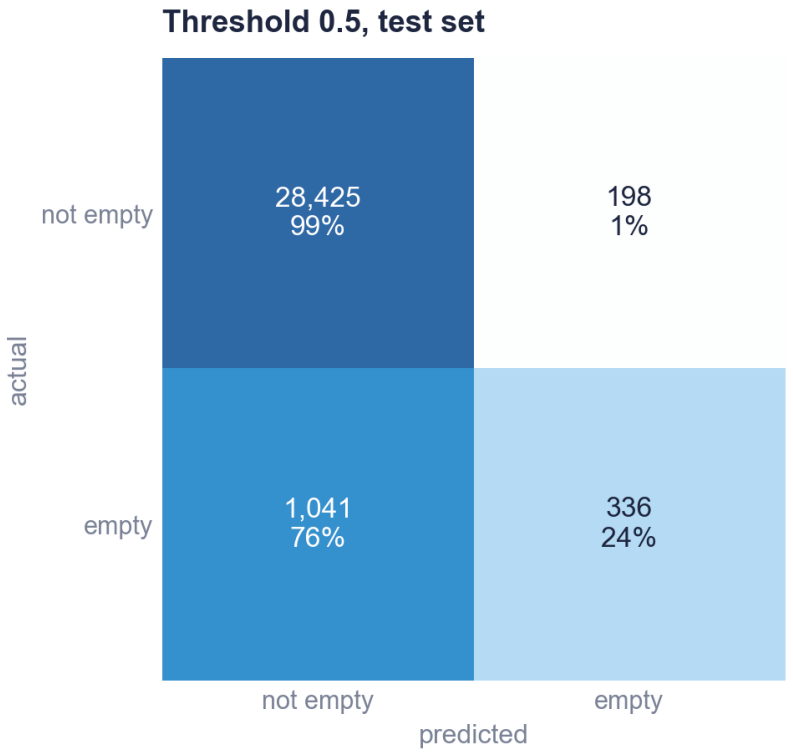


1 In brackets: share of all station-days in each group. Right: logarithmic vertical scale; in blue the station-days counted as empty (target = 1).

At the default threshold the model is cautious; for trucks we rank instead

Left: precision and recall of XGBoost for every decision threshold. Right: confusion matrix at the default threshold of 0.5.[1]

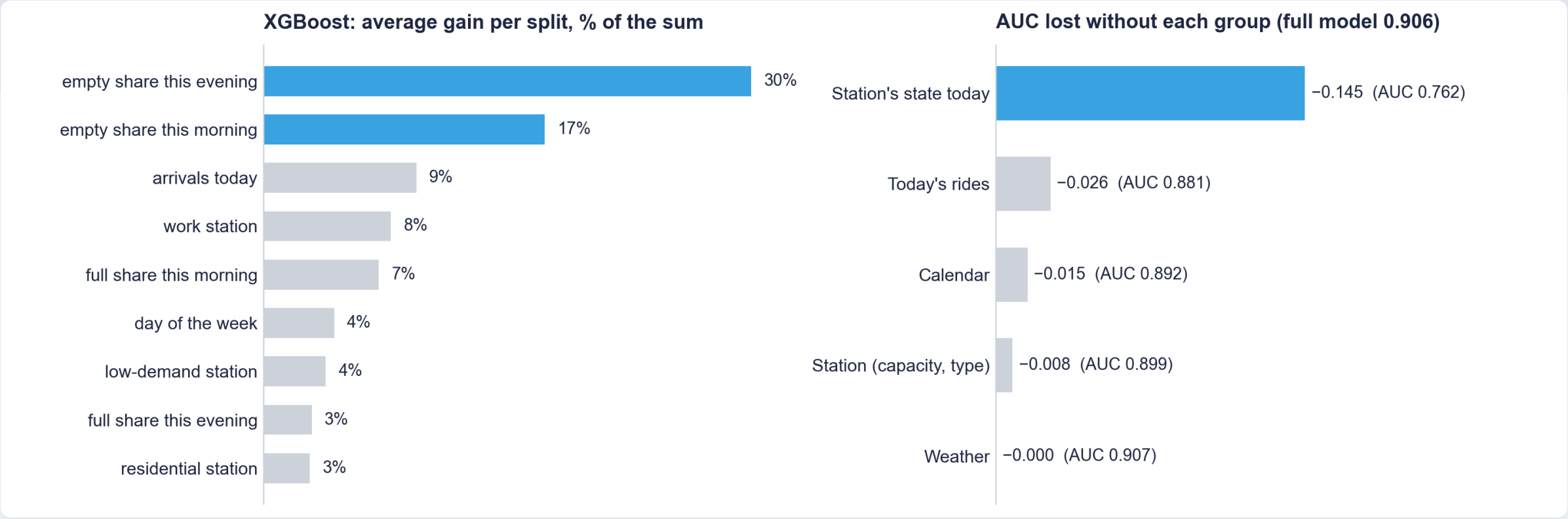




1 Precision = listed stations that are really empty. Recall = empty stations that are listed. Test set of 30,000 station-days.

Today's state of the station drives the prediction

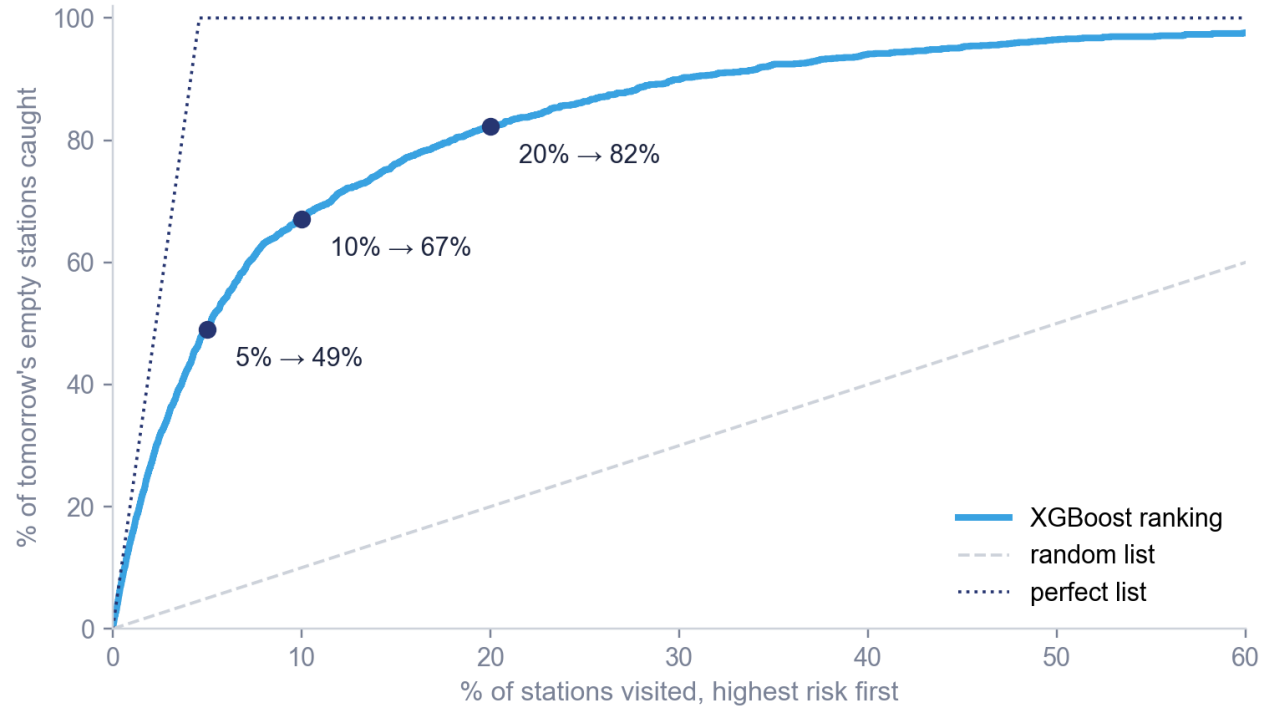
Left: XGBoost importance (average gain per split).[1] Right: test AUC lost when a group of variables is removed and the model retrained.[2]

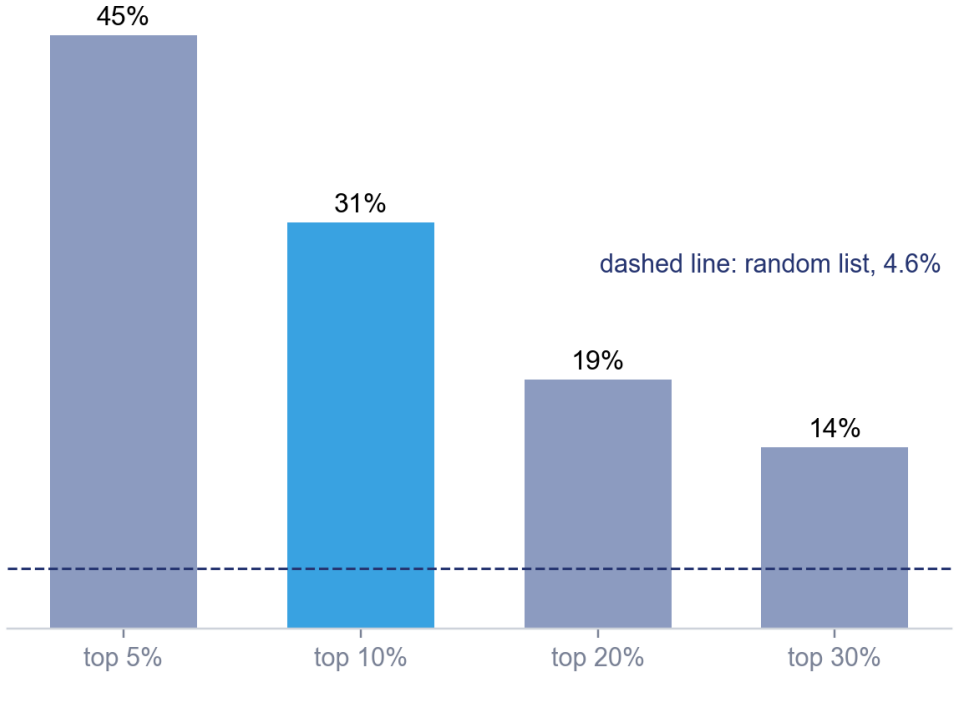


1 get_score(importance_type='gain'): average improvement of the loss each time the variable is used in a split; top 9 shown.
2 Ablation: XGBoost retrained without the group, same settings, on the same 30,000 test station-days.

On the random test set, a 10% list catches two thirds of empty stations

Left: share of empty station-days caught when visiting in order of predicted risk. Right: hit rate by list size. Random test set of 30,000 station-days.[1]

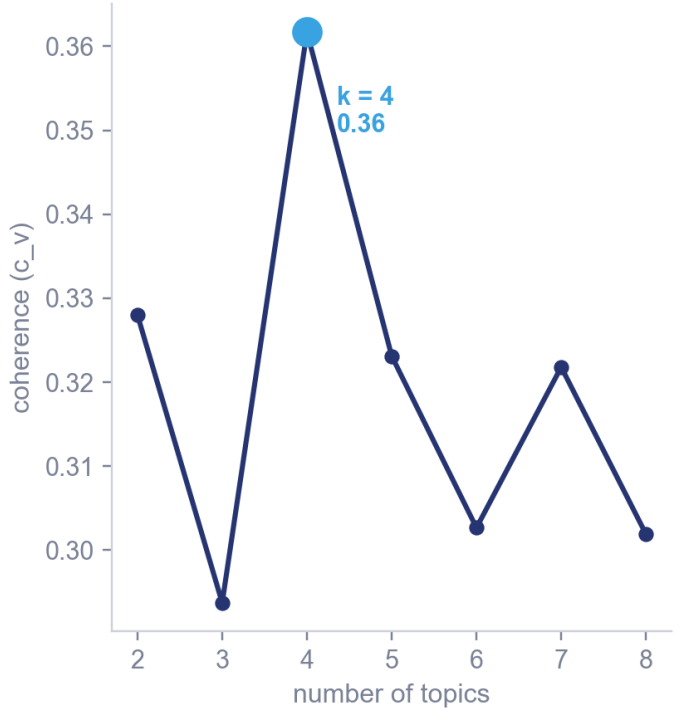


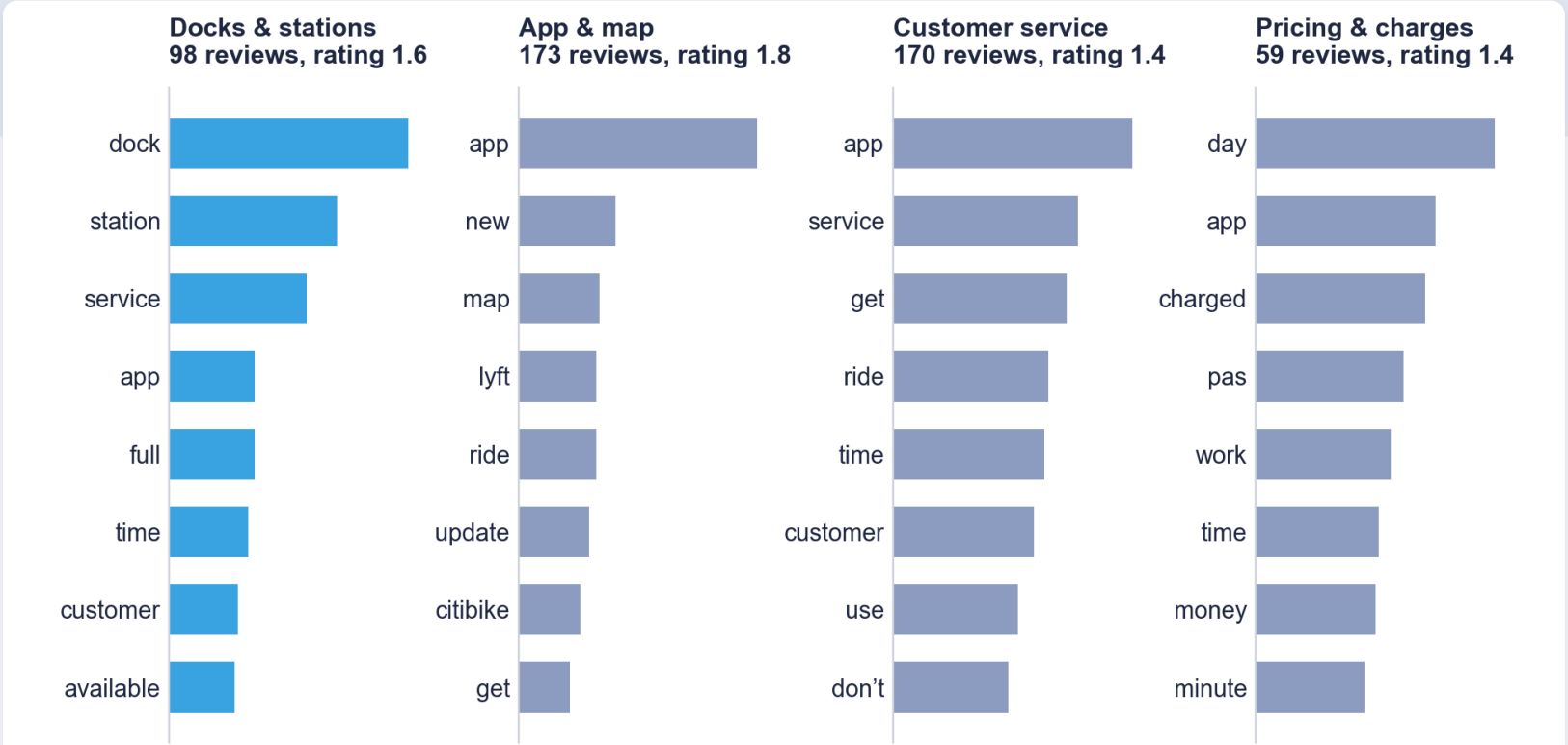


1 Hit rate = share of listed station-days that are empty ≥ 1 hour of 7–10am the next day. Perfect list: all empty stations first.

Topic model: four topics chosen by coherence

LDA topic model on 500 App Store reviews: number of topics chosen by coherence, and the most likely words of each topic.[1]





1 gensim LdaModel and CoherenceModel. Main topic of a review = topic with the highest probability. Words are lemmatised ("pas" comes from "pass"). Rating: 1 to 5 stars.